

OCR Further Mathematics
Additional Pure Mathematics
Mark Scheme
AS and A Level
Further Mathematics A - H235, H245



Exam Papers Practice

The latest topic questions for the year 2024 / 2025+ exams.
To be used by all students preparing for OCR AS and A Level Further Mathematics.
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Question			Answer/Indicative content	Marks	Part marks and guidance	
1	a		The section $x = 0$, i.e. $z = f(0, y)$ drawn The section $y = 0$, i.e. $z = f(x, 0)$ drawn	B1 (A03.4) B1 (A03.4) [2]	U-shaped parabola in y-z plane (axes labelled) $\cap$-shaped parabola in x-z plane (axes labelled)	
	b		Suggestion $z = ay^2 - bx^2$ $(-0.25 \leq x \leq 0.25, -0.3 < y < 0.3)$ $z = 0.27$ when $x = 0, y = (\pm) 0.3$ $\Rightarrow a = 3$ $z = -0.4$ when $y = 0, x = (\pm) 0.25 \Rightarrow b = 6.4$	M1* (A03.3) A1 (A01.1) M1*dep (A03.4) A1 (A01.1) A1 (A02.2a) [5]	M for quadratic (only) terms in both x and y Details, including signs of coefficients (Domain not required) Use of "boundary" conditions to evaluate a, b (at least one fully attempted)	
			Total	7		
2	a		For constant sequence, set $2 = \frac{k}{6+2} \text{ (from } u_2 = u_1)$ $\Rightarrow k = 16$	M1 (A01.1a) A1 (A01.1) [2]		
	b		$u_3 = \frac{k}{6 + \frac{k}{8}} = \frac{8k}{48 + k}$ equated to $u_1 = 2$ Solving for $k \Rightarrow 96 + 2k = 8k$ $\Rightarrow k = 16$ But this is the condition for $\{u_n\}$ constant, so the sequence is never periodic with period 2	M1 (A03.1a) A1 (A01.1) A1 (A02.3) [3]	u_3 correct (simplified) and set = 2 Correct conclusion stated with supporting reason	



Question			Answer/Indicative content	Marks	Part marks and guidance	
	c		$u_4 = \frac{k}{6 + \frac{8k}{48 + k}} = \frac{k(k+48)}{288 + 14k},$ $u_5 = \frac{k}{6 + \frac{k(k+48)}{288 + 14k}} = \frac{k(14k+288)}{k^2 + 132k + 1728}$ For $\{u_n\}$ periodic, period 4: $u_5 = u_1$ $\Rightarrow 14k^2 + 288k = 2k^2 + 264k + 3456$ $12(k^2 + 2k - 288) = 0 \Rightarrow (k - 16)(k + 18) = 0$ $k = -18$	M1 (AO1.1) A1 (AO1.1) M1 (AO3.1a) M1 (AO1.1) A1 (AO2.2a) [5]	u_4 and u_5 attempted in terms of k At least u_4 correct (simplified) u_5 must have been worked out and an algebraic expression equated to 2 Solving a quadratic eqn. in k Correct (single) answer only	
			Total	10		



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Question			Answer/Indicative content	Marks	Part marks and guidance	
3	a		Aux. Eqn. is $m^2 - 5m + 4 = 0 \Rightarrow m = 1, 4$ $\Rightarrow$ Gen. Soln. is $H_n = A + B \times 4^n$ $H_0 = 3 \Rightarrow 3 = A + B$ and $H_1 = 7 \Rightarrow 7 = A + 4B$ $\Rightarrow A = \frac{5}{3}, B = \frac{4}{3}$ givi $H_n = \frac{1}{3}(5 + 4^{n+1})$ ng	M1 (A01.1) A1 (A01.1) M1 (A01.1) M1 (A01.1) A1 (A01.1) [5]	Use of initial terms Solving simultaneous eqns. in A and B	
	b	i	0, 1, 4, 5, 6, 9	B1 (A01.1) [1]	In any order	
		ii	For $n = 0, 1, 2, \dots$ $4^n = 4, 6, 4, 6, \dots \pmod{10}$ $\Rightarrow 5 + 4^n = 9, 11, 9, 11, \dots \pmod{10}$ $\equiv 9, 21, 9, 21, \dots \pmod{10}$ $\Rightarrow \frac{1}{3}(5 + 4^n) \equiv 3, 7, 3, 7, \dots \pmod{10}$ since $\text{hcf}(3, 10) = 1$ and 3, 7 are quadratic non-residues modulo 10, so $H_n \neq$ a square	M1 (A03.1a) A1 (A01.1) M1 (A01.1) A1 (A02.1) A1 (A02.2a) A1 (A02.4) [6]	OR M1 for noting the terms of the sequence mod 10 so that final A1 can be earned also	
			Total	12		



Question			Answer/Indicative content	Marks	Part marks and guidance	
4	a		$I_n = \int_0^{\frac{1}{2}\pi} \cos^n x \, dx = \int_0^{\frac{1}{2}\pi} \cos^{n-1} x \cos x \, dx$ $= [\cos^{n-1} x \sin x] - \int_0^{\frac{1}{2}\pi} (n-1) \cos^{n-2} x (-\sin x) \sin x \, dx$ $= 0 + (n-1) \int_0^{\frac{1}{2}\pi} \cos^{n-2} x (1 - \cos^2 x) \, dx = (n-1) \{I_{n-2} - I_n\}$ $\Rightarrow n I_n = (n-1) I_{n-2}$	M1 (AO1.1) A1 (AO1.1) M1 (AO3.1a) A1 (AO1.1) [4]	Correct splitting and attempted use of 'parts' Correct unsimplified Setting up into I_k terms only AG legitimately obtained	
	b		$I_0 = \frac{1}{2}\pi \text{ evaluated and}$ $I_n = \left(\frac{n-1}{n} \right) I_{n-2} \text{ used}$ repeatedly gives $I_1 = \frac{1}{2}\pi, I_2 = \frac{1}{4}\pi, I_3 = \frac{3}{8}\pi, I_4 = \frac{1}{2}\pi, I_5 = \frac{15}{32}\pi$	M1 (AO1.1) A1 (AO1.1) [2]	Note: candidates may choose to evaluate $I_2 = \frac{1}{4}\pi$ directly as a starting-point	
	c		$\int_0^{\frac{1}{2}\pi} \cos^6 x \sin^2 x \, dx = \{I_6 - I_8\}$ $= \frac{5}{32}\pi - \frac{35}{256}\pi = \frac{5}{256}\pi$	M1 (AO3.1a) A1 (AO2.2a) [2]	Converting into I_k terms and using the previous results from part above NB Missing factor of 2 and / or incorrect sign penalised only here at the end FT $I_6 - I_8 = \frac{1}{7} I_8 = \frac{1}{7}$ (answer from above part)	
			Total	8		



Question			Answer/Indicative content	Marks	Part marks and guidance	
5	a		Closed... YES since the 'product' of 2 complex numbers is a third	B1 (AO2.4) [1]	'Yes' with valid reason	
	b		$[(a + ib) \diamond (c + id)] \diamond (e + if)$ $= [ac + i(b + ad)] \diamond (e + if)$ $= ace + i(b + ad + acf)$ $(a + ib) \diamond [(c + id) \diamond (e + if)]$ $= (a + ib) \diamond [ce + i(d + cf)]$ $= ace + i(b + a[d + cf]) =$	M1 (AO1.2) A1 (AO1.1) M1 (AO2.1) A1 (AO1.1) [4]	Attempt at first of $(pq)r$ or $p(qr)$, correct to here Attempt at the other, correct to here Must be convincingly shown equal to first answer	
	c		$1 + i0$ is the identity since $(1 + i0) \diamond (a + ib) =$ $1 \cdot a + i(0 + 1b) = (a + ib) \diamond$ $(1 + i0)$	B1 (AO2.2a) B1 (AO2.4) [2]	Demonstrated (condone one-side only)	
	d		$(a + ib) \diamond (c + id) = 1 \Rightarrow ac + i(b + ad) = 1$ $\Rightarrow c = \frac{1}{a} \text{ and } d = \frac{-b}{a}$ S is the set of complex numbers with non-zero real part (The 'exclusions' are those complex numbers with zero real part; i.e. the strictly imaginary numbers)	M1 (AO3.1a) A1 (AO2.2a) B1 (AO3.2a) [3]	Test for an inverse of a general element Both components of inverse element found Explained	
			Total	10		



Question		Answer/Indicative content	Marks	Part marks and guidance	
6	a	Differentiating $x^2 + y^2 + z^2 = xyz - 1$ partially w.r.t. either x or y x): $2x + 2z \frac{\partial z}{\partial x} = xy \frac{\partial z}{\partial x} + yz$ ❶ y): $2y + 2z \frac{\partial z}{\partial y} = xy \frac{\partial z}{\partial y} + xz$ ❷ $x \cdot ❶ + y \cdot ❷ \Rightarrow 2(x^2 + y^2) + 2z \left(x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} \right) = xy \left(x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} \right) + 2xyz$ Now $x^2 + y^2 + z^2 = xyz - 1 \Rightarrow xyz - x^2 - y^2 = z^2 + 1$ giving $(2z - xy) \left(x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} \right) = 2(1 + z^2)$	M1 (A01.1) A1 (A01.1) A1 (A01.1) M1 (A03.1a) M1 (A01.1) A1 (A02.2a) [6]	Attempt with LHS correct First correct Correct or FT $x \leftrightarrow y$ (by symmetry) $\Rightarrow (2z - xy) \left(x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} \right) = 2(xyz - x^2 - y^2)$ AG legitimately obtained	
	b	If both $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ are zero then LHS = 0 while RHS ≥ 2 (i.e. > 0) ($\Rightarrow \Leftarrow$)	M1 (A02.1) A1 (A02.4) [2]	Considering conditions for stationary points Contradiction justified	
		Total	8	Exam Papers Practice	

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Question	Answer/Indicative content	Marks	Part marks and guidance
7	$x = 3a + 1 = 11b + 5$ <u>mod 11</u> : $3a \equiv 4 \equiv 15 \Rightarrow a \equiv 5 \pmod{11}$ (since $\text{hcf}(3, 11) = 1$) $(a = 11c + 5)$ $x = 33c + 16$ $2x = 66c + 32 = 17d + 5$ <u>mod 17</u> : $15c \equiv -27$ $\Rightarrow 5c \equiv -9 \equiv 8 \equiv 25$ $\Rightarrow c \equiv 5 \pmod{17}$ (since $\text{hcf}(5, 17) = 1$) Then $c = 17e + 5 \Rightarrow x = 561e + 181$ or $x = 181 \pmod{561}$ Stating at least one of either $\text{hcf}(3, 11) = 1$ or $\text{hcf}(5, 17) = 1$ Alternative Method 1 $2x \equiv 5 \equiv 22 \Rightarrow x \equiv 11 \pmod{17} \equiv -6 \pmod{17}$ Then $x \equiv -6 \pmod{11}$ $x \equiv -6 \pmod{187} = 181$ since $\text{hcf}(11, 17) = 1$ Note that 181 is already $\equiv 1 \pmod{3}$ Since there is a unique solution $\pmod{3 \times 11 \times 17 = 561}$ the solution is $x \equiv 181 \pmod{561}$ Alternative Method 2 The Chinese Remainder Theorem (CRT) There is a unique solution $\pmod{M = 3 \times 11 \times 17 = 561}$ $2x \equiv 5 \equiv 22 \Rightarrow x \equiv 11 \pmod{17}$ $a_1 = 1, M_1 = \frac{M}{3} = 187,$ R_1 is the reciprocal of $M_1 \pmod{3} = 1$ $a_2 = 5, M_2 = \frac{M}{11} = 51,$	M1 (AO1.1) M1 (AO2.1) A1 (AO3.1a) M1 (AO2.1) A1 (2.2a) B1 (AO2.4) Alternative Method 1 1 M1 M1 A1 B1 M1 A1 Alternative Method 2 2 B1 M1 M1 A1 M1 A1 [6]	Equating first two equations Working mod 11 or via $b = 3c + 1$ (working mod 3) Substituting into 3 rd equation and working mod 17 Alternative Method 1 OR continue as above: $x = 3a + 1 = 187b + 181$ (as 3, 11, 17 are pairwise co-prime) Alternative Method 2 Since $187 \equiv 1 \pmod{3}$ anyway For any one set of terms correct Since $51 \equiv 7 \pmod{11}$ and $7 \times 8 = 56 \equiv 1 \pmod{11}$ All three sets of terms correct. Since $33 \equiv -1 \equiv 16 \pmod{17}$ CRT employed correctly (in principle) BC



Question			Answer/Indicative content	Marks	Part marks and guidance	
			R_2 is the reciprocal of M_2 $\text{mod } 11 = 8$ $a_3 = 11, M_3 = \frac{M}{17} = 33,$ R_3 is the reciprocal of M_3 $\text{mod } 17 = 16$ Then $x \equiv a_1 M_1 R_1 + a_2 M_2 R_2 + a_3 M_3 R_3 \pmod{561}$ $= 1 \times 187 \times 1 + 5 \times 51 \times 8 + 11 \times 33 \times 16 = 8035 = 181 \pmod{561}$			
			Total	6		
8	a		$\mathbf{p} \times \mathbf{q} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 1 & 5 \\ 1 & -1 & 1 \end{vmatrix} = \begin{pmatrix} 6 \\ 3 \\ -3 \end{pmatrix}$ $= 3 \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} \Rightarrow t = -1$	M1 (A03.1a) A1 (A02.2a) [2]	Possibly BC Correct vector product; t correct	
	b		Vol. $OABC = \frac{1}{6} \mathbf{p} \times \mathbf{q} \cdot \mathbf{r} = \frac{1}{6} \left \begin{pmatrix} 6 \\ 3 \\ -3 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 1 \\ t \end{pmatrix} \right $ $= \frac{1}{6} 12 + 3 - 3t $ Solving $5 - t = 26$ and / or $t - 5 = 26$ $\Rightarrow t = -21$ or 31	B1 (A01.1) M1 (A01.1) M1 (A03.1a) A1 (A02.2a) [4]	A correct scalar triple product involving t Correct formula attempted Method may be implied by one correct t	
			Total	6		



Question			Answer/Indicative content	Marks	Part marks and guidance	
9	a		$\{e, a, a^2\}$ $\{e, b\} \{e, ab\} \{e, a^2b\}$	B1 (A01.1) B1 (A01.1) B1 (A01.1) B1 (A01.1) [4]	Ignore inclusion of $\{e\}$, G	
	b		G is isomorphic to the symmetric group of three elements i.e. D_3 (the symmetries of the Δ) or S_3 (the permutations of 3 symbols)	B1 (A02.5) [1]	Accept "the non-cyclic group of order 6"	
			Total	5		
10			$41723 = 10 \times 16^3 + 2 \times 16^2 + 15 \times 16 + 11$ $= A2FB_{16}$	M1 (A01.1) A1 (A01.1) A1 (A01.2) [3]	Writing in terms of powers of 16; correct	
			Total	3		



Question		Answer/Indicative content	Marks	Part marks and guidance	
11	a	$u_1^2 - 3 = -2 < 0$ so result is true for $n = 1$ Assume that $u_k^2 - 3 < 0$ for some $k \geq 1$ The $u_{k+1}^2 - 3 = \left(\frac{2u_k + 3}{u_k + 2}\right)^2 - 3$ n $= \frac{(2u_k + 3)^2 - 3(u_k + 2)^2}{(u_k + 2)^2}$ $= \frac{4u_k^2 + 12u_k + 9 - 3u_k^2 - 12u_k - 12}{(u_k + 2)^2} = \frac{u_k^2 - 3}{(u_k + 2)^2} < 0$ Explanation that 1 st case true and k^{th} case true $\Rightarrow (k + 1)^{\text{th}}$ case true gives the result for all positive integers n	B1 (AO1.1) M1 (AO1.1) M1 (AO2.1) M1 (AO1.1) A1 (AO2.2a) E1 (AO2.4) [6]	Considering the $(k + 1)^{\text{th}}$ case Common denominator Must be noted that it is negative	
	b	$u_{n+1} - u_n = \frac{2u_n + 3}{u_n + 2} - \frac{u_n(u_n + 2)}{u_n + 2} = \frac{3 - u_n^2}{u_n + 2}$ Explaining that $u_{n+1} > u_n$ from (a)'s result	M1 (AO3.1a) A1 (AO1.1) B1 (AO2.2a) [3]		
	c	$\frac{3 - a^2}{a + 2} = 0 \Rightarrow a = \pm\sqrt{3}$ from (b) $u_{n+1} = \sqrt{3}$ $> u_n$ so	M1 (AO3.1a) A1 (AO1.1) [2]	Special Case B1 for correct limit with invalid or missing reasoning	
	d	Monotonic (or strictly) increasing, convergent	B1 (AO2.5) [1]		
		Total	12		



Question			Answer/Indicative content	Marks	Part marks and guidance	
12	a		By “Fermat’s Little Theorem”, since 5 is prime.	B1 (AO1.2) [1]	i.e. <i>FLT</i>	
	b		If n is even, then n^5 is also even If n is odd, then n^5 is also odd so that $n^5 \equiv n \pmod{2}$ Since $\text{hcf}(2, 5) = 1$ it follows that $n^5 \equiv n \pmod{2 \times 5 = 10}$	M1 (AO2.1) A1 (AO1.1) A1 (AO2.4) [3]	Conclusion with supporting reason	
	c	i	Need to show that $n^5 \equiv n \pmod{3}$ $\text{mod } 3, 0^5 = 0 \equiv 0, 1^5 = 1 \equiv 1, 2^5 = 32 \equiv 2$ Because $n^5 \equiv n \pmod{10}$, then $n^5 - n$ is divisible by 10 Since $\text{hcf}(3, 10) = 1$ it follows that $n^5 \equiv n \pmod{3 \times 10 = 30}$ i.e. $n^5 - n$ is divisible by 30 $\forall n \in \mathbb{N}$ Alternative method $n^3 \equiv n \pmod{3}$ by <i>FLT</i> Then $n^5 = n^3 \cdot n^2 \equiv n \cdot n^2 = n^3 \equiv n \pmod{3}$ Because $n^5 \equiv n \pmod{10}$, then $n^5 - n$ is divisible by 10 Since $\text{hcf}(3, 10) = 1$ it follows that $n^5 \equiv n \pmod{3 \times 10 = 30}$ i.e. $n^5 - n$ is divisible by 30 $\forall n \in \mathbb{N}$	B1 (AO3.1a) M1 (AO2.1) A1 (AO1.1) M1 (AO3.1a) A1 (AO2.4) B1 M1 A1 M1 A1 [5]	Running through all possibilities All noted correctly Conclusion with supporting reason Conclusion with supporting reason	
		ii	No – since, when $n = 2, n^5 - n = 30$	B1 (AO2.3) [1]	Producing a counter-example	



Question			Answer/Indicative content	Marks	Part marks and guidance	
			Total	10		
13	a		2, 3, 4 and 6 because these are the factors of 12 (by Lagrange's Theorem)	B1 (AO1.1) B1 (AO2.4) [2]	Ignore 1 or 12 if they appear	
	b		{3, 9, 27}	B1 (AO1.1) [1]		
	c		27	B1 (AO 1.1) [1]		
	d		18, $18^2 = 324 = 12 \pmod{39}$, $(18^3 = 21) \dots$ $18^4 = 12^2 = 27 \pmod{39}$ so order 4	M1 (AO1.1) A1 (AO2.1) [2]	For method of searching by finding powers	
	e		$3 \equiv 3, 42, 81, \dots$ First square-root of 3 is 9 Second square-root of 3 is - $9 \equiv 30$	M1 (AO3.1a) A1 (AO2.2a) A1 (AO3.2a) [3]	Method for searching for squares $\equiv 3 \pmod{39}$ Cannot just state -9	
	f		Identity 27 in G does not map to identity 1 in H	B1 (AO2.4) [1]	Or any clear equivalent statement	
			Total	10		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
14	a	i	$z = \cosh x$ (<i>catenary</i> curve) through (0, 1), symmetrical about z -axis	B1 (AO3.4) [1]	Penalise line extending outside $x \in [-\ln 20, \ln 20]$ Axes must be x - and z -axes and labelled thus	
		ii	$L = \int \sqrt{1 + \left(\frac{dz}{dx}\right)^2} dx$ used $\frac{dz}{dx} = \sinh x$ with $= \int \sqrt{1 + \sinh^2 x} dx$ or $\int \cosh x dx$ with correct limits $= 19.95$ or via $2 \sinh(\ln 20)$ < 20 so YES (design requirement met)	M1 (AO3.1b) A1 (AO1.1) A1 (AO1.1) A1 (AO3.5a) [4]	Condone use of y instead of z May have a factor of 2 if limits (0, $\ln 20$) used BC or via correct integration From cao with stated conclusion	
	b	i	$y = \frac{1}{\cosh x}$ sketched (must be through (0, 1) and symmetrical about y -axis) Axes must be x - and y -axes and labelled thus	B1 (AO3.4) [1]	Allow ft for reciprocal of previous function (provided all positive) Condone line extending outside $x \in [-\ln 20, \ln 20]$ only if already penalised in (a) (i)	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$\frac{dy}{dx} = \frac{-\sinh x}{\cosh^2 x}$ $A = 2\pi \int y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$ used with y and $\frac{dy}{dx} = \dots$ $= 2\pi \int_{-\ln 20}^{\ln 20} \frac{1}{\cosh x} \sqrt{1 + \frac{\sinh^2 x}{\cosh^4 x}} dx$ Use of $\sinh^2 x \equiv \cosh^2 x - 1$ $= 4\pi \int_0^{\ln 20} \frac{1}{\cosh^3 x} \sqrt{\cosh^4 x + \cosh^2 x - 1} dx$	B1 (AO1.1) M1 (AO1.1) A1 (AO1.1) M1 (AO3.1a) A1 (AO2.3) [5]	Correct derivative (with correct sign) Condone (incorrect sign)-squared AG Must show clearly how the limits give $k = 4$ and how the integrand is as shown	
		iii	$\int$ above (given) is $4\pi \times$ 1.564 710 33... = 19.7 so YES (design requirement met)	B1 (AO1.1) B1 (AO3.5a) [2]	BC correct to ≥ 3 s.f. (19.662 73 to 5 d.p.) Cao Correct conclusion must be stated	
			Total	13		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
15	a	i	$c \times b = -a$	B1 (AO1.1) [1]		
		ii	$a \times (b \times c) = a \times a$ $= 0$	M1 (AO1.1) A1 (AO1.2) [2]	Must be a vector zero	
		iii	$a \cdot (b \times c) = a \cdot a$ $= a ^2 = 9$	M1 (AO1.1) A1 (AO2.2a) [2]		
	b	i	Using $b \times c = a$ OA is parallel to BC OR OA is normal to the plane OBC	B1 (AO 2.2a) [1]		
		ii	Using (a) (iii): tetrahedron $OABC$ has volume 1.5 (cubic units) OR noting that O, A, B, C are not co-planar OR (FT from (b) (i)) Prism with base OBC and height OA has volume 9	B1 (AO 2.2a) [1]		
			Total	7		

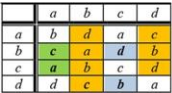


Question			Answer/Indicative content	Marks	Part marks and guidance	
16	a		$I_n + I_{n-2} = \int_0^1 \frac{x^{n-2}(1+x^2)}{1+x^2} dx = \int_0^1 x^{n-2} dx = \left[\frac{x^{n-1}}{n-1} \right]_0^1 = \frac{1}{n-1}$	M1 (AO1.1 1.1) M1 A1 (AO1.1) [3]	Combining as a single integral (added numerators) Factorising and cancelling AG legitimately obtained	
	b	i	$I_0 = \frac{1}{4} \pi$ $I_2 = 1 - \frac{1}{4} \pi, \quad I_4 = \frac{1}{3} - I_2 = \frac{1}{4} \pi - \frac{2}{3},$ $I_6 = \frac{1}{5} - I_4 = \frac{13}{15} - \frac{1}{4} \pi$ $I_8 = \frac{1}{7} - I_6 = \frac{1}{4} \pi - \frac{26}{105}, \quad I_{10} = \frac{1}{9} - I_8 = \dots$ $\text{or } I_{10} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \frac{1}{4} \pi = \frac{263}{315} - \frac{1}{4} \pi$	B1 (AO1.1) M1 (AO1.1) M1 (AO1.1) A1 (AO1.1) [4]	BC Use of the Reduction Formula repeatedly ($\geq \times 3$) All the way to I_{10} Correct final answer	
		ii	$I_{10} > 0 \Rightarrow \frac{1}{4} \pi < \frac{263}{315}$ $\Rightarrow \pi < \frac{263}{315} \times 4 = \frac{1052}{315} = 3 \frac{107}{315}$	M1 (AO3.1a) A1 (2.2a) [2]	Clear statement and consequence noted AG from clear demonstration	
			Total	9		



Question		Answer/Indicative content	Marks	Part marks and guidance	
17	a	$f_x = 4xy^7 + 15x^4y^4 - 40x^7y$ and $f_y = 14x^2y^6 + 12x^5y^3 - 5x^8$ Then $xf_x + yf_y = 4x^2y^7 + 15x^5y^4 - 40x^8y + 14x^2y^7 + 12x^5y^4 - 5x^8y = 18x^2y^7 + 27x^5y^4 - 45x^8y = 9f$	M1 (AO1.1) A1 A1 (AO1.1) (AO1.1) M1 (AO2.1) A1 (AO1.1) [5]	Clear attempt at partial differentiation $n = 9$ must be clearly stated	
	b	$f_{xx} = 4y^7 + 60x^3y^4 - 280x^6y$ and $f_{xy} = 28xy^6 + 60x^4y^3 - 40x^7$ Then $xf_{xx} + yf_{xy} = 4xy^7 + 60x^4y^4 - 280x^7y + 28xy^7 + 60x^4y^4 - 40x^7y = 32xy^7 + 120x^4y^4 - 320x^7y = 8f_x$ Alternative method Differentiate. (a)'s result w.r.t. x : $xf_{xx} + f_x + yf_{yx} = nf_x$ M1 A1 $f_{yx} = f_{xy}$ (by the Mixed Derivative Theorem) B1 $\Rightarrow xf_{xx} + yf_{xy} = (n - 1)f_x$ A1	B1 B1 (AO1.1) (AO1.1) M1 (AO 2.1) A1 (AO 1.1) [4]		
		Total	9		



Question			Answer/Indicative content	Marks	Part marks and guidance	
18	a		 Column 1 Column 3 Any 3rd R or C All correct	B1 (AO 3.1a) B1 (AO 1.1) B1 (AO 2.2a) B1 (AO 1.1) [4]		
	b		No – since c is the “left identity” but there is no “right identity” Or there is no identity column Or $(ad)b = cb = b$ <u>but</u> $a(db) = ac = a$ so op. not associative	B1 (AO 2.4) [1]	Any valid reason (vague or unsupported claims are not acceptable)	
			Total	5		



Question		Answer/Indicative content	Marks	Part marks and guidance	
19	a	DR $2(p-2)^{p-2} = 2(-2)^{p-2} \pmod{p}$ $= -(-2)^{p-1}$ $\equiv -1 \pmod{p} \text{ by FLT}$ since $\text{hcf}((-2), p) = 1$ Alt. I DR Since p and $(p-2)$ are consecutive odds, $\text{hcf}(p-2, p) = 1$ and so $(p-2)^{p-1} \equiv 1 \pmod{p}$ by FLT $\Rightarrow (p-2)(p-2)^{p-2} \equiv 1 \pmod{p}$ $\Rightarrow (-2)(p-2)^{p-2} \equiv 1 \pmod{p}$ $\Rightarrow 2(p-2)^{p-2} \equiv -1 \pmod{p}$ Alt. II DR $(p-2)^{p-2} = p^{p-2} + (p-2) \binom{p-2}{1} p^{p-3} (-2) + \binom{p-2}{2} (p^{p-4}) (-2)^2 + \dots$ $\dots + \binom{p-2}{p-4} (p^3) (-2)^{p-4} + (p-2)p(-2)^{p-3} + (-2)^{p-2} \equiv (-2)^{p-2} \pmod{p}$ Then $2(p-2)^{p-2} = -(-2)^{p-1} \equiv 1 \pmod{p}$ and $(-2)^{p-1} \equiv 1 \pmod{p}$ by FLT $\Rightarrow \text{result since } \text{hcf}(-2, p) = 1$	M1 A1(A02.1 1.2) A1(A02.4) E1(A02.2a) E1 B1 M1 A1 M1 A1 A1 E1 [4]	AG correctly reasoned By the <i>Binomial Theorem</i> (since each term apart from the last has a factor of p) Statement only is fine Can also argue via $(p-2)$ but there's more to justify <u>Examiner's Comments</u> There is little doubt that candidates found this to be the hardest question on the paper and it elicited a lot of poorly-constructed answers and conceptual uncertainties as a result. Most of those who did make a serious attempt at least recognised that Fermat's little theorem (FLT) was needed at some stage. Unnecessarily, though not incorrectly, the use of the binomial theorem was frequently used, but a lot of attempts suffered from the impulse to start dividing, or working with an unspecified	



Question			Answer/Indicative content	Marks	Part marks and guidance	
					inverse / reciprocal of $(p - 2)$ that then proved an insurmountable problem. As mentioned earlier on in the introductory overview section, those candidates who started trying to divide were rarely triumphant. Successful candidates were those who pointed out that, $\text{mod } p$, $(p - 2) \equiv -2$, since multiples of the modulus can always be removed (or introduced) without harm. Some of these candidates justified this step by deploying the binomial theorem, but this is an unnecessary distraction as it should be obvious that this is so. Having reduced $(p - 2)^{p-2}$ to $(-2)^{p-2}$, it is then simple to note that the exponent must be odd (as p is an odd prime) and so this is now equal to $-(2^{p-2})$. At which point, the extra factor of 2 in the question leads to the use of FLT and the result comes out after only three or four short lines of working. However, the use of FLT in the form $a^{p-1} \equiv 1 \pmod{p}$ does require the checking of the appropriate condition (on this occasion that $\text{hcf}(2, p) = 1$) and many otherwise successful solutions missed out on the mark for this.	

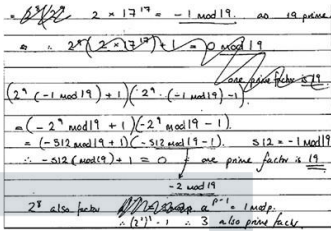


Question		Answer/Indicative content	Marks	Part marks and guidance	
	b	DR $N \equiv 2 \times (1)^{34} - \text{mod } 2^{15}$ 3, $\equiv 2 \times 1 - 2 = 0$ since $2^{\text{odd}} = 2$ (mod 3) and $2^{\text{even}} = 1 \pmod{3}$ $2 \times 34^{34} - 2^{15} = 2^{35} \times 17^{34} - 2^{15} = 2^{15}(2^{20} \times 17^{34} - 1)$ $(2^{20} \times 17^{34} - 1) = (2^{10} \times 17^{17} - 1)(2^{10} \times 17^{17} + 1)$ Using (a), $2 \times 17^{17} = -1 \pmod{19}$ so first bracket $= -2^9 - 1 = -513 = -19 \times 27 = 0 \pmod{19}$ and N is a multiple of 19 Alt. (for factor 19) $2 \times 34^{34} - 2^{15} = 2^{35} \times 17^{34} - 2^{15} = 2^{15}(2^{20} \times 17^{34} - 1) = 2^{15}B$ mod 19, $2^{18} = 1$ by <i>FLT</i> $B = 2^2 \times 17^{34} - 1 = (2 \times 17^{17} - 1)(2 \times 17^{17} + 1)$ Using (a), second bracket is divisible by 19 and N is a multiple of 19 Alt. (for factor 19) $2 \times 34^{34} - 2^{15} = 2^{33}(2 \times 17^{34} - 2^{15}) = 2^{33} \times 17^{17} \cdot 2^{17} - 2^{15} = 2^{33}(-1)^2 - 2^{15} \pmod{19}$ $= 2^{33} - 2^{15} = 2^{15}(2^{18} - 1) \pmod{19}$ $2^{18} - 1 \equiv 0 \pmod{19}$ by <i>FLT</i> since $\text{hcf}(2, 19) = 1$ and N is a multiple of 19	M1(A03.1a) A1(A02.4) B1(A03.1a) M1(A02.5) M1(A03.1a) M1(A02.4) A1(A03.2a) B1 M1 M1 M1 A1 [7] B1 M1 M1 M1 A1 [7]	Working mod 3 Fully justified that $3 \mid N$ Indices work to factor out the 2^{15} Use of the <i>difference-of-two-squares</i> factorisation Using (a)'s result Working mod 19 in one or both brackets Valid conclusion Indices work to factor out the 2^{15} (since 19 is prime and $\text{hcf}(2, 19) = 1$) Use of the <i>difference-of-two-squares</i> factorisation Using (a)'s result Valid conclusion Indices work to obtain a part (a) expression Using (a)'s result Working mod 19 with remaining numerical term(s) Use of <i>FLT</i> or calculator Valid conclusion (<i>FLT</i> justified or correctly demonstrated numerical work) Note that $2^{18} - 1 = 262\,143$	

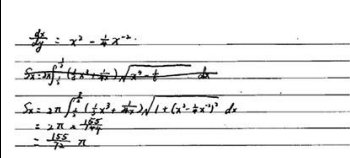


Question			Answer/Indicative content	Marks	Part marks and guidance	
			NB $N = 2^{15} \times 3 \times 19 \times 389 \times \dots$		$= 19 \times 13\,797$ and $2^{33} - 2^{15} = 8\,589\,901\,824 = 19 \times 452\,100\,096$	
					<u>Examiner's Comments</u> Many candidates that had struggled with (a) seemed to have little appetite for this and made no attempt here, but it is good exam technique to recognise that one can use given results (as here in (a)) without having proved them. Among those who did produce an uneasy response, far too many looked to test for divisibility by 17 (or 34, as if 36 were the prime being referred to). The most sensible starting point (unfortunately taken by few candidates) when asked to look for odd factors is to remove as many even factors (i.e. powers of 2) as possible before proceeding. Extracting, and then ignoring, the 2^{15} then leaves you with $2^{20} \times 17^{34} - 1$. Those who made this obvious step were then the beneficiaries of an easy mark for using the 'difference of two squares factorisation, which led to $(2^{10} \times 17^{17} - 1)(2^{10} \times 17^{17} + 1)$ and $p = 19$ follows. Candidates taking these steps were then able to continue to show that 19 is one of the required odd factors of N . Only a few candidates spotted the obvious factor of 3, even among those	



Question			Answer/Indicative content	Marks	Part marks and guidance
					who made progress towards the 19; arguments here tended to vary from far too concise to excessive detail. In simple terms, again it is far better with the maximum number of 2s gone, then to note that $34 \equiv 1 \pmod{3}$ and $2^2 \equiv 1 \pmod{3}$ also. Exemplar 6  Exemplar 6 is an excerpt from a solution to Q8(b) that nicely illustrates some very interesting points. Firstly, by factoring out the powers of 2 in two stages a mark has been lost by overlooking that 2^7 has been extracted from each of the two brackets. Nonetheless, the usefulness of a 'difference of two squares factorisation' has still been spotted and this has been left unaffected by the incorrect power of 2 (since it can now safely be ignored). Having taken two steps into this approach, it is now clear that what remains relates back to part (a)'s result, with $p - 2 = 17$ emerging. This candidate went on to score 4 of the 5 marks available for the factor of 19, losing only the mark noted.
			Total	11	



Question		Answer/Indicative content	Marks	Part marks and guidance	
20	a	$\frac{dy}{dx} = x^2 - \frac{1}{4x^2} \Rightarrow 1 + \left(\frac{dy}{dx}\right)^2 = 1 + \left(x^2 - \frac{1}{4x^2}\right)^2$ $\text{or } \left(x^2 + \frac{1}{4x^2}\right)^2$ $A = 2\pi \int \left(\frac{x^3}{3} + \frac{1}{4x}\right) \left(x^2 + \frac{1}{4x^2}\right) dx$ or $2\pi \int \left(\frac{x^5}{3} + \frac{x}{3} + \frac{1}{16x^3}\right) dx$ $= 2\pi \left[\frac{x^6}{18} + \frac{x^2}{6} - \frac{1}{32x^2} \right]_{\frac{1}{2}}^{\frac{3}{2}} = 2\pi \left(\frac{1145}{576} - \frac{-95}{576} \right) = \frac{155}{72} \pi$	M1 A1(A01.1) A1(A01.1a1.1) A1(A01.1) [4]	Not necessarily as a perfect square Correct integrand so at any stage (any correct form, including involving a square-root) BC or from correct integration	<u>Examiner's Comments</u> A large majority of candidates scored 3 or 4 here. Candidates began by quoting the correct surface area integral formula and working out the integrand, but then candidates seemed to be split equally among those who completed the integration process by hand and those who used a calculator to identify the correct rational multiple of π. Both were acceptable, though the second approach was probably quicker and less prone to small numerical errors. Exemplar 5 



Question			Answer/Indicative content	Marks	Part marks and guidance	
					Exemplar 5 nicely illustrates the correct application of the 'principle of least effort' and serves as an excellent exemplar for any of the three parts of Q7. The correct derivative has been written down (despite calling it dx/dy for no obvious reason; such irrelevant 'typos' are generally ignored within working until such a point where they affect the correctness or validity of the ensuing mathematical working). The surface area formula has then been quoted in context (i.e. with the relevant content for this question), followed by a clear statement of the integral to be evaluated. To cap off this excellently concise solution, this candidate has then resorted to their calculator to feed them the correct output value for the integral. This is a case in point regarding the general remark made earlier about when it is appropriate to use a calculator and this candidate has judged it to perfection. Although one might expect good candidates at this level to be able to deal quite easily with an integral of this kind (once one has obtained a square-root of a perfect square), an understanding of the assessment should reveal that this is work that does not need to be done that way, as what is being Y545 assesses does not	



Question			Answer/Indicative content	Marks	Part marks and guidance
					include basic A Level integration methods. In the absence of a “detailed reasoning...” demand or similar command word, candidates are quite at liberty to presume that their solution need not be any more algebraic than this one displayed here. In addition, many candidates who did attempt the integration lost time in so doing and often lost the single mark for which they were striving through some often minor numerical slip along the way.
	b		Line is $y = \frac{3}{4}x + \frac{1}{6}$ and cuts the x-axis at $(-\frac{2}{9}, 0)$ DR Distance $(-\frac{2}{9}, 0)$ to $(\frac{3}{2}, \frac{31}{24})$ is $\sqrt{\left(\frac{31}{18}\right)^2 + \left(\frac{31}{24}\right)^2} = \frac{155}{72}$ Distance $(-\frac{2}{9}, 0)$ to $(\frac{1}{2}, \frac{13}{24})$ is $\sqrt{\left(\frac{13}{18}\right)^2 + \left(\frac{13}{24}\right)^2} = \frac{65}{72}$ Curved SA required is Alt. Line is $y = \frac{3}{4}x + \frac{1}{6}$	B1(A03.3) M1(A03.4) A1(A01.1) A1(A01.1) B1 M1 A1 A1 [4]	Cutting-point on x-axis BC Attempt at either of the two slant heights (or use of calculus) Both correct Cao (possibly BC) Attempt at surface area integral with y and “ds” Correct integrand in any form (ignore limits for now) Cao (possibly BC)



Question			Answer/Indicative content	Marks	Part marks and guidance	
			$SA = 2\pi \int \left(\frac{3}{4}x + \frac{1}{6}\right)\left(1 + \left(\frac{3}{4}\right)^2\right) dx$ $= \frac{55}{24}\pi$		<u>Examiner's Comments</u> Almost all candidates approached this using integration very successfully, through finding the equation of the line joining P and Q and then using the same process as for (a). A few lost marks by being careless with this line equation. Again, calculators were frequently used to find the multiple of π required for the answer.	



Question		Answer/Indicative content	Marks	Part marks and guidance	
	c	Mean value $M = \frac{1}{\frac{3}{2} - \frac{1}{2} - 0.5} \int_{\frac{1}{3}}^{\frac{1.5}{3}} \left(\frac{x^3}{3} + \frac{1}{4x} \right) dx$ of y is $= \left[\frac{x^4}{12} + \frac{1}{4} \ln x \right]_{0.5}^{1.5} = \frac{5}{12} + \frac{1}{4} \ln 3$ or 0.691... Curved SA = $2\pi rh$ (with r cylinder = ym, $h = 1$) $= 2\pi \left(\frac{5}{12} + \frac{1}{4} \ln 3 \right) = 4.3437 \text{ to } 4 \text{ d.p.}$	M1(A03.3) A1(A01.1) M1 A1(A03.4 1.1) [4]	Attempt to find the mean value of y over this interval BC Or by integration Allow exact answer and condone awrt 4.344 (i.e. to 4 s.f.) <u>Examiner's Comments</u> Many more candidates actually turned to an entirely calculus approach for the first part before going decimal for the final answer. A lot of mistakes were made with the integral of $\frac{1}{4x}$, but otherwise this was also handled very capably with around half of all candidates gaining all 4 marks.	
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Question		Answer/Indicative content	Marks	Part marks and guidance	
21	a	$\mathbf{p} \cdot \mathbf{q} \times \mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -17 \\ -14 \end{pmatrix} \text{ or } \begin{vmatrix} 1 & 3 & 2 \\ 2 & 1 & -4 \\ 3 & -1 & 5 \end{vmatrix} = -75$ $\mathbf{p} \times (\mathbf{q} \times \mathbf{r}) = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \times \begin{pmatrix} 1 \\ -17 \\ -14 \end{pmatrix} = \begin{pmatrix} 23 \\ 17 \\ -19 \end{pmatrix}$ $(\mathbf{p} \times \mathbf{q}) \times \mathbf{r} = \begin{pmatrix} -5 \\ 10 \\ -5 \end{pmatrix} \times \begin{pmatrix} 2 \\ -4 \\ 5 \end{pmatrix} = \begin{pmatrix} 30 \\ 15 \\ 0 \end{pmatrix}$	M1(AO1.1a) A1(AO1.1) M1(AO1.1a) A1(AO1.1) M1(AO1.1a) A1(AO1.1) [6]	One vector product and a scalar product Or via the determinant method Two vector products, in correct order Two vector products, in correct order	
	b	No, since $\mathbf{p} \times (\mathbf{q} \times \mathbf{r}) \neq (\mathbf{p} \times \mathbf{q}) \times \mathbf{r}$	B1(AO2.2a) [1]	Correct answer with reason <u>Examiner's Comments</u> Technically, part (a) provided one example for which $\mathbf{p} \times (\mathbf{q} \times \mathbf{r}) \neq (\mathbf{p} \times \mathbf{q}) \times \mathbf{r}$, but this is sufficient to conclude that the vector product is not an associative operation. Most candidates duly noted this.	



Question		Answer/Indicative content	Marks	Part marks and guidance	
	c	$b \times c = n$ is normal / perpendicular to the plane containing b, c (or B, C) Then $a \times n$ is perpendicular to this normal and hence in the plane of b, c so that $a \times (b \times c)$ is of the form $\lambda b + \mu c$ for scalar constants λ and μ.	M1(A03.1a) A1(A03.2a) [2]	Consideration of fact that the vector product gives a vector perpendicular to the two original vectors Fully explained ("planes" not essential; directions suffice) <u>Examiner's Comments</u> Although there is a little more to it than this, essentially all that was being looked for here was some explanation that $b \times c$ is a vector perpendicular to both b and c, and that $a \times (b \times c)$ is perpendicular to this normal. Whether this is 'in' some plane or merely 'parallel to' it requires a little more care in one's description and many candidates bravely made a fuller description in order to justify the final given form.	

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Question			Answer/Indicative content	Marks	Part marks and guidance	
	d		$(a \times b) \times c = -c \times (a \times b)$ $= -\{(c \cdot b) a - (c \cdot a) b\}$ $= (c \cdot a) b - (c \cdot b) a$ Alt. NB $(a \times b) \times c = -c \times (a \times b) = -c \times -(b \times a) = c \times (b \times a) = (c \cdot a) b - (c \cdot b) a$	M1(A03.1a) M1(A02.1) A1(A02.2a) M1 A1 [3]	Anti-commutative property used Attempted use of given form, with $a \rightarrow c, b \rightarrow a, c \rightarrow b$ Correct, in required form (order of dot products may be reversed, of course) <u>Examiner's Comments</u> Unfortunately, rather a lot of candidates seemed to just guess here, with the most popular choice being exactly the one quoted in (*). All that was wanted was an application of the anti-commutative property of the vector product followed by a re-ordering of the vectors contained within (*) to match.	
			Total	12		



Question			Answer/Indicative content	Marks	Part marks and guidance																										
22	a		$11^2 = 121 \equiv 41 \pmod{80}$ so $41 \in S$ $11 \times 41 = 451 \equiv 51 \pmod{80}$ so $51 \in S$ $41^2 \equiv 1 \pmod{80}$ so $1 \in S$ <table><tr><td>$\times_{80}$</td><td>1</td><td>11</td><td>41</td><td>51</td></tr><tr><td>1</td><td>1</td><td>11</td><td>41</td><td>51</td></tr><tr><td>11</td><td>11</td><td>41</td><td>51</td><td>1</td></tr><tr><td>41</td><td>41</td><td>51</td><td>1</td><td>11</td></tr><tr><td>51</td><td>51</td><td>1</td><td>11</td><td>41</td></tr></table>	$\times_{80}$	1	11	41	51	1	1	11	41	51	11	11	41	51	1	41	41	51	1	11	51	51	1	11	41	B1(A03.1a) B1(A02.5) B1(A02.2a) M1(A01.1a) A1(1.1)[5]	B1 for each other element found (possibly tabulated) 4×4 table attempted (with at least R1 and C1 correct) using the correct 4 elements All correct <u>Examiner's Comments</u> The key here is to keep finding powers of 11 until one arrives at the identity (and a repeating cycle of elements following it). Looking to the following parts of the question, it should have been clear that G was going to be of order 4. The vast majority of candidates seemed perfectly at ease here and could be given all 5 marks on offer. Of the remaining candidates, the most popular response seemed to incorporate the integers from 1 to 11 before stopping, earning only the one mark for identifying the obvious identity '1'.	
$\times_{80}$	1	11	41	51																											
1	1	11	41	51																											
11	11	41	51	1																											
41	41	51	1	11																											
51	51	1	11	41																											
	b		Choice of a suitable method, such as Listing orders of elements: 1, 4, 2, 4 in G and 1, 2, 2, 2 in H OR Stating that G is the cyclic group of order 4 while H is the Klein-4 group OR Stating that G has an element of order 4 while H	M1(A02.1) A1(A02.4) [2]	MUST be for two groups of order 4 Using knowledge of small-order groups: M1 for clear indication that there are only two groups of order 4, A1 for the details																										



Question			Answer/Indicative content	Marks	Part marks and guidance	
			has all (non-identity) elements of order 2		<u>Examiner's Comments</u> There were various approaches used by candidates here. The most popular one involved noting the differences between the orders of elements (while H consists of all self-inverse elements, of order 2, G has at least one element of order 4). Others pointed out that G was an example of the cyclic group of order 4, C_4 , while H was an example of K_4 , the Klein group of order 4. To earn full marks here it was important for candidates to make clear and relevant comments. For instance, ' G has a generator, H doesn't' is too vague and needs amplifying with a bit more detail as to what this means. Comments like 'The orders of the elements in H are..., while the orders of the elements in G are...' need to be supported with correct orders to count.  AFL There is a significant discrepancy concerning the treatment (and description) of the properties of a group's identity element. In general, it is taken to have order 1, even though it satisfies the criteria which define a self-inverse element (of order 2). In this question, the orders of the elements of G are 1, 2, 4, 4	



Question			Answer/Indicative content	Marks	Part marks and guidance	
					and the orders of the elements of H are 1, 2, 2, 2. The reader may be reassured on this front, however; as a result of the different ways of referring to the identity and its properties, claims made by candidates are only considered for the <i>non-identity</i> elements of groups. Exemplar 3 <i>eg Every element in H has order 2 whereas 2 of the elements in G have order 4 (11/51)</i> The main purpose of this exemplar is to indicate that it is really not necessary to write lengthy responses. This candidate identifies a suitable method for explaining a fundamental difference in group structure and then does so in an admirably succinct way. Bearing in mind the immediately preceding remarks about the identity, the claim that <i>all</i> of H's elements are self-inverse is taken to refer to the non-identity elements. In comparing this property with G's elements, even this candidate has gone slightly further than necessary, since one actually only needs to note that there is one element in G that is not self-inverse to be credited.	



Question			Answer/Indicative content	Marks	Part marks and guidance	
	c	i	G has proper subgroup $\{1, 41\}$ only H has proper subgroups $\{1, 9\}$, $\{1, 31\}$ and $\{1, 39\}$	B1(A03.1a) M1(A01.2) A1(A02.2a) [3]	Ignore statement of $\{1\}$ and group throughout Listing any two correct subgroups of order 2 All three and no extras <u>Examiner's Comments</u> The vast majority of candidates were able to identify the proper subgroups (all of order 2) of both groups.  AFL There is another significant discrepancy relating to the topic of groups that arises here and it is not quite so easy to tidy up definitively, as there are differences of opinion even among standard group theory texts. Teachers and students can be reassured however that the examiners are aware of this and (as with the identity issue mentioned above) fully intend to ignore the inclusion or exclusion of the sets $\{e\}$ and G in these sorts of questions. There are two main standpoints. The first is that a <i>proper</i> subgroup does not include the identity subgroup and the whole group itself; these being referred to as the <i>trivial</i> subgroups, since every group obviously has these two subgroups. The second view is that a proper subset H of a set G, written $H \subset G$ (as opposed to $H \subseteq G$), is not equal to the set G and	



Question			Answer/Indicative content	Marks	Part marks and guidance	
					hence G is, by this definition, not a <i>proper</i> subgroup of itself. However, this definition does not preclude the identity subgroup as being described as ' <i>proper</i> ' and this can cause confusion for candidates.	
		ii	G and H must have different structures since they have differing numbers of (proper) subgroups	B1(A02.4) [1]	(Must have different numbers of subgroups in (c)(i)) <u>Examiner's Comments</u> Candidates who had any kind of sensible answer to part (c)(i) realised that, in order to be isomorphic, the two groups would necessarily have to contain equal numbers of subgroups (of the same kind / order). Exemplar 4 <i>all elements in H have the order 2 and can generate a subgroup itself whereas only one element in G can.</i> Exemplar 4 demonstrates that it is possible to make completely correct statements that still do not achieve the mark. In this case, the candidate's reasons might have earned marks in part (b), but here the question requires the solver to refer back to the proper subgroups found in (c)(i).	
			Total	11		



Question		Answer/Indicative content	Marks	Part marks and guidance	
23	a	PS is $T_n = -29$ CS from $\lambda^2 + 2 = 0$, $\lambda = \pm i\sqrt{2}$, is $T_n = A(i\sqrt{2})^n + B(-i\sqrt{2})^n$ $T_n = A(i\sqrt{2})^n + B(-i\sqrt{2})^n$ General Solution is Use of $T_0 = -27$ and $T_1 = 27$ to create simultaneous equation in A, B Two correct equations: $A + B = 2$ and $A - B = -28i\sqrt{2}$ Solving attempt at two equations: $A = 1 - 14i\sqrt{2}$ and $B = 1 + 14i\sqrt{2}$ $T_n = (-14i\sqrt{2})(i\sqrt{2})^n + (1 + 14i\sqrt{2})(-i\sqrt{2})^n - 29$ i.e. Alt. version PS is $T_n = -29$ CS from $\lambda^2 + 2 = 0$, $T_n = (\sqrt{2})^n (C \cos \frac{n\pi}{4} + D \sin \frac{n\pi}{4})$ $\lambda = \pm i\sqrt{2}$, is $(\sqrt{2})^n (C \cos \frac{n\pi}{4} + D \sin \frac{n\pi}{4}) = 29$ General Solution is Use of $T_0 = -27$ and $T_1 = 27$ to create simultaneous equation in C, D Two correct equations: $C - 29 = -27$ and $\sqrt{2} D - 29 = 27$ Solving attempt at two	B1(A01.1) M1(A01.1a) A1(A01.1) B1(A01.2) M1(A01.1a) A1(A01.1) M1(A01.1a) A1(A01.1) B1 M1 A1 B1 M1 A1 A1	Particular Solution Complementary Solution attempted from auxiliary equation; correct FT GS = CS (with 2 arbitrary constants) + PS (with none) Or unsimplified equivalents cao Particular Solution Complementary Solution attempted from auxiliary equation; correct FT GS = CS (with 2 arbitrary constants) + PS (with none) Unsimplified	



Question			Answer/Indicative content	Marks	Part marks and guidance	
			equations: $C = 2$ and $D = 28\sqrt{2}$ $T_n = (\sqrt{2})^n (2\cos\frac{an}{2} + 28\sqrt{2}\sin\frac{an}{2}) - 29$ i.e.	A1 [8]	cao	<u>Examiner's Comments</u> Although a question such as this (involving the solution of a second-order recurrence system (linear, with constant coefficients)) is relatively routine in principle, in practice it was relatively demanding due to the appearance of either complex numbers or trigonometric terms. Very few candidates elected to pursue the second of these two solution-paths and this was almost certainly for the best; candidates opting for trigonometric terms invariably did not identify the correct fundamental angle (simply $\frac{1}{2}\pi$) and performed poorly overall. Otherwise, the high level of success on this question was a fine testimony to the capabilities of the candidates' background knowledge in this respect, with around half of the candidates managing at least 7 of the 8 marks for this part. It is an interesting point to note that part of this question required the solution of a pair of simultaneous equations and that there was no reason



Question			Answer/Indicative content	Marks	Part marks and guidance
					<i>not</i> to resort to the calculator, most of which cope readily with complex numbers. A very small number of candidates (very shrewdly) recognised that the nature of the given recurrence relationship means that the odd- and even-numbered terms are independent of each other. Giving the answers for each of these two sub-sequences was usually completely successfully and for the record is below. $T_n = \begin{cases} 56 \times (-2)^{\frac{1}{2}(n+1)} & n \text{ odd} \\ 2 \times (-2)^{\frac{1}{2}n} & n \text{ even} \end{cases}$
	b		Substituting $n = 20 \Rightarrow$ $T_{20} = (1 - 14i\sqrt{2}) \times 1024 + (1 + 14i\sqrt{2}) \times 1024 - 29$ $= 2048 - 29 = 2019$	M1(A02.1) A1(A01.1) [2]	Or by applying the given recurrence relation (e.g. BC) <u>Examiner's Comments</u> Even those candidates who hadn't been able to sort out part (a) entirely satisfactorily could still work part (b) of the question, since T_{20} can be calculated directly from the sequence's definition, possibly even by means of a suitable calculator.
			Total	10	



Question		Answer/Indicative content	Marks	Part marks and guidance	
24	a	$7x \equiv 6 \equiv 25 \equiv 44 \equiv 63 \dots$ $\Rightarrow x \equiv 9 \pmod{19}$ Alt. $7^{-1} \pmod{19}$ is 11 and multiplying through by this $\Rightarrow x \equiv 9 \pmod{19}$	M1(A01.1) A1(A01.1) M1 A1 [2]	Adding 19s to RHS to reach a multiple of 7 Must be a general result (i.e. not just $x = 9$) Must be a general result (i.e. not just $x = 9$) No requirement to justify by $\text{hcf}(7, 19) = 1$ e.g. <u>Examiner's Comments</u> This was another early question where a large majority of the candidates scored successfully. Of those that didn't pick up both marks, many of them lost one mark by arriving at the answer " $x = 9$ ", which appears to give the impression that there IS only the one answer, rather than infinitely many (of which 9 is a representative). " $x = 9 \pmod{19}$ " is acceptable, although the use of the correct notation in the proper way is preferred. Although not allocated a mark here in Q3, candidates using the 'division' approach (see comments in the introductory overview above) could (in similar future questions) be required to justify this move by noting the co-primality of some relevant factor with 19. Most of those who <i>did</i> mention that (for instance) $\text{hcf}(7, 19) = 1$ did so in order to note that a (unique)	

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Question			Answer/Indicative content	Marks	Part marks and guidance	
					solution did indeed exist, rather than to justify the 'division' step.	
	b		$x \equiv 3 \pmod{4} \Rightarrow x$ is odd while $x \equiv 4 \pmod{6} \Rightarrow x$ is even Statement that there are no solutions Alt. $h = \text{hcf}(4, 6) = 2$ and solutions exist provided $h \mid (4 - 3)$ But $2 \nmid 1$ hence no solutions	M1(A03.1a) A1(A02.1) M1 A1 [2]	Consideration of parity (i.e. mod 2) or equivalent For valid reason Considering h and " $b - a$ " Conclusion with justification <u>Examiner's Comments</u> The intention here was that candidates would note that the first statement necessarily requires x to be odd, while it must be even in the second, i.e. there is a clear contradiction. In summer 2019 responses, this was matched for frequency with responses quoting a result that is not actually on the syllabus, namely that for $x \equiv a \pmod{m_1}$ and $x \equiv b \pmod{m_2}$, solutions exist only when $\text{hcf}(m_1, m_2) \mid (b - a)$. While this is a perfectly acceptable approach to take, there were still quite a number who tried to quote it, but couldn't quite recall the details accurately and ended up without marks. It also raises the issue that, while <i>external</i> results may be valid and useful, the questions are actually constructed on the basis of what is on the syllabus and that all questions can be answered without recourse to additional material.	
			Total	4		



Question		Answer/Indicative content	Marks	Part marks and guidance	
25	a	$f_x = 2x \sin y - 2y \sin x$ $f_{xx} = 2 \sin y - 2y \cos x$ $f_{xy} = f_{yx} = 2x \cos y - 2 \sin x$	$f_y = x^2 \cos y + 2 \cos x$ $f_{yy} = -x^2 \sin y$	B1B1 (A01.1a 1.1) B1B1 (A01.1 2.5) B1(A01.1) [5]	Both must be written down, or stated equal <u>Examiner's Comments</u> Here candidates that have grasped the basic principles of partial differentiation scored highly and almost all candidates picked up the 5 marks. Almost all of the successful candidates worked out both f_{xy} and f_{yx} separately, with very few just stating that they are equal (due to a result known as the <i>Mixed Derivative theorem</i> for 'well-behaved' functions).

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Question			Answer/Indicative content	Marks	Part marks and guidance	
	b	i	When $x = y = \frac{1}{2}\pi$, $f_x = f_y = 0$ $\Rightarrow$ stationary point Visible check / calculation that $z = \left(\frac{1}{2}\pi\right)^2 \times 1 + 2 \times \left(\frac{1}{2}\pi\right) \times 0 = \frac{1}{4}\pi^2$	M1(A01.1a) A1(A02.2a) B1(A02.1) [3]	Allow statement only <u>Examiner's Comments</u> This part of the question was also well handled by the overwhelming majority of candidates, however almost half lost one of the 3 marks by failing to address all necessary matters. The two that were most prone to oversight being the mark for noting that $f_x = 0$ and $f_y = 0$ for a stationary point (which needed also to be demonstrated and not just stated) and, in particular, the mark for verifying the value of z at the point given. A small number of candidates mistakenly thought they were required to <i>solve</i> $f_x = 0$ and $f_y = 0$ (or, occasionally, $f_x = f_y$).	
		ii	$H = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix}$ attempted $H = \begin{vmatrix} 2 & -2 \\ -2 & -\frac{1}{4}\pi^2 \end{vmatrix} = -\frac{1}{2}\pi^2 - 4$ (or -8.935-ish) $\Rightarrow$ Saddle-point since $H < 0$	M1(A01.1) A1(A02.1) B1(A02.2a) [3]	Considering (the determinant of) the Hessian matrix, or equivalent Correctly calculated value of H FT conclusion from $H < 0$ <u>Examiner's Comments</u> Again, this part of the question was dealt with very capably by almost all candidates, though again	



Question			Answer/Indicative content	Marks	Part marks and guidance
					one mark was frequently lost due to a lack of care in either the numerical values (exact or decimal) of the four second partial derivatives or the evaluation of the Hessian's determinant. Exemplar 2a $H = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix} = \begin{pmatrix} 2\sin y - 2\cos x & 2x\cos y - 2\sin x \\ 2x\cos y - 2\sin x & -2x^2\cos y - 2\sin y \end{pmatrix}$ $f_{xx} = 2\sin y - 2\cos x$ $f_{xy} = 2x\cos y - 2\sin x$ $f_{yy} = -2x^2\cos y - 2\sin y$ $x = \frac{\pi}{2}, y = \frac{\pi}{2}$ $H = \begin{pmatrix} 2\sin(\frac{\pi}{2}) - 2\cos(\frac{\pi}{2}) & 2(\frac{\pi}{2})\cos(\frac{\pi}{2}) - 2\sin(\frac{\pi}{2}) \\ 2(\frac{\pi}{2})\cos(\frac{\pi}{2}) - 2\sin(\frac{\pi}{2}) & -2(\frac{\pi}{2})^2\cos(\frac{\pi}{2}) - 2\sin(\frac{\pi}{2}) \end{pmatrix} = \begin{pmatrix} 2 & -2 \\ -2 & -2 \end{pmatrix}$ $ H = 2(-2) - 4 = -8 < 0$ $\therefore$ it is a saddle-point. Exemplar 2b $\text{at } (\frac{\pi}{2}, \frac{\pi}{2}, \frac{\pi}{2}), H = \begin{pmatrix} 2 & -2 \\ -2 & -2 \end{pmatrix}$ $ H = 2(-2) - 4 = -8 < 0$ $\therefore$ it is a saddle-point. Exemplar 2a has made life very hard by attempting to work the determinant of the Hessian <i>algebraically</i>. For Exemplar 2b, a simple statement of the relevant values (worked out by calculator, presumably) suffices and then the following working and conclusion are made to look very straightforward
			Total	11	



Question			Answer/Indicative content	Marks	Part marks and guidance
26			$u_2 = 3$ $u_3 = 0.8$ $u_4 = 0.6, u_5 = 2$ and $u_6 = 5$ Explanation that $u_0 = u_5$ and $u_1 = u_6 \Rightarrow$ periodicity, period 5	M1(A01.1a) M1(A01.1) A1(A01.1) E1(A02.4) [4]	Use of the given r.r. (with at least u_2 correct) Repeated use of the given r.r. (with at least u_3 correct) u_4, u_5 & u_6 correct (since a second-order r.r.) <u>Examiner's Comments</u> This was intended to be a fairly gentle introduction to the paper. Even without the aid of a modern calculator (which can very speedily generate sequences of this kind), it is very easy to calculate the next few terms. Indeed, just working as far as u_6 earned 3 of the 4 marks. However, around a quarter of all candidates stopped at the very first repeated term (i.e. u_5), thereby losing 2 marks. This represents a misunderstanding, since the given recurrence relationship is <i>second-order</i>, each term relying on the two previous terms and for this reason, candidates need to find a repeated pair of consecutive terms. Finally, as this was a 'proof' question, a final acknowledgement of this was needed; of the 70% of candidates scoring 3 or 4 marks on this question, the numbers were split fairly evenly between those who did and those who didn't. Exemplar 1a



Question			Answer/Indicative content	Marks	Part marks and guidance
					$U_n = \frac{1+U_{n-1}}{U_{n-2}}$ $U_0 = 2, U_1 = 5$ $U_2 = \frac{1+5}{2} = 3$ $U_3 = \frac{1+3}{5} = \frac{4}{5}$ $U_4 = \frac{1+\frac{4}{5}}{3} = \frac{9}{15} = \frac{3}{5}$ $U_5 = \frac{1+\frac{3}{5}}{\frac{4}{5}} = 2$ The sequence is periodic: 2, 5, 3, $\frac{4}{5}$, $\frac{3}{5}$ and has period 5. Exemplar 1b $U_n = \frac{1}{U_{n-2}} + \frac{U_{n-1}}{U_{n-2}}$ $U_0 = 2, U_1 = 5, U_2 = 3, U_3 = \frac{4}{5}, U_4 = \frac{3}{5}, U_5 = 2, U_6 = 5$ Hence returns to initial values of u_0 and u_1 at u_5 and u_6. The sequence 'will then repeat this periodicity' period 5. Exemplar 1a stops at U_5 (the first repeating term) and this scored 2 of the 4 marks. Exemplar 1b has stopped just one term later (scoring the third mark). Then, despite having no explicit statement that the sequence is defined recursively as a function of two consecutive terms, there is a clear implication that the required result follows from the fact that $\{U_5, U_6\} = \{U_0, U_1\}$ and this is sufficient to earn the fourth mark for the requested 'proof' (on this occasion, a fairly informal one).
			Total	4	

Copyright © 2024 Exam Papers Practice If $t > \frac{9}{4}$, S_t has a (local) minimum since $\det(H) > 0$	A1 (AO 1.1) M1 (AO 3.1a)	Attempt to find Hessian in term of t correct
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Question			Answer/Indicative content	Marks	Part marks and guidance	
28		a	$I_1 = \int_1^5 x\sqrt{2+x^2} dx$ $= \left[\frac{1}{3} (2+x^2)^{\frac{3}{2}} \right]_1^5$ $= \frac{1}{3} (81\sqrt{3} - 3\sqrt{3}) = 26\sqrt{3}$	M1 (AO 1.1) A1 (AO 1.1) A1 (AO 2.2a) [3]	Substitute $n = 1$ and attempt integral by substitution or inspection For $k(2+x^2)^{\frac{3}{2}}$, any non-zero k Correctly gained from correct k and use of (1, 5) limits	
		b	$I_n = \int_1^5 x^{n-1} \cdot x\sqrt{2+x^2} dx$ $= x^{n-1} \cdot \frac{1}{3} (2+x^2)^{\frac{3}{2}} - \int (n-1)x^{n-2} \cdot \frac{1}{3} (2+x^2)^{\frac{3}{2}} dx$ $3I_n = 5^{n-1} \cdot 81\sqrt{3} - 3\sqrt{3} - (n-1) \int_1^5 x^{n-2} (2+x^2)^{\frac{3}{2}} dx$ $= 5^{n-1} \cdot 81\sqrt{3} - 3\sqrt{3} - (n-1) \{2I_{n-2} + I_n\}$ $\Rightarrow (3+n-1)I_n = 3\sqrt{3} (27 \times 5^{n-1} - 1) - 2(n-1)I_{n-2} \Rightarrow \text{Result}$	M1 (AO 1.1a) A1A1 (AO 1.1, 1.1) M1 (AO 2.1) A1 (AO 1.1) A1 (AO 1.1) [6]	Correct splitting and attempt to integrate by parts One for each term correctly integrated (ignore limits until the end) Splitting the second integral's terms, ready to get I 's Correct second integral in terms of I 's AG Legitimately obtained	
		c	Use of $n = 3$ in Reduction Formula: $5I_3 = 3\sqrt{3} (27 \times 25 - 1) - 2(2)I_1$ $I_3 = \frac{1}{5} (2022\sqrt{3} - 104\sqrt{3}) = \frac{1918}{5}\sqrt{3}$ Use of $n = 5$ in Reduction Formula: $7I_5 = 3\sqrt{3} (27 \times 625 - 1) - 2(4)I_3$ $I_5 = \frac{1}{7} (50622\sqrt{3} - 8 \times \frac{1918}{5}\sqrt{3}) = \frac{237766}{35}\sqrt{3}$	M1 (AO 1.1a) A1 (AO 1.1) M1 (AO 1.1) A1 (AO 2.2a) [4]	Candidates may work downwards	
			Total	13		



Question			Answer/Indicative content	Marks	Part marks and guidance	
29		a	$\mathbf{T} = \mathbf{a} \times \mathbf{F}_1 = \begin{pmatrix} 3 \\ 4 \\ 1 \end{pmatrix} \times \begin{pmatrix} 6 \\ 7 \\ -3 \end{pmatrix} = \begin{pmatrix} -19 \\ 15 \\ -3 \end{pmatrix}$	M1 (AO 3.4) A1 (AO 1.1) [2]	Attempt at cross-product, soi BC	
		b	$\mathbf{b} \times \mathbf{F}_2 = \begin{pmatrix} 3 \\ 5 \\ 1 \end{pmatrix} \times \begin{pmatrix} -7 \\ -10 \\ 2 \end{pmatrix} = \begin{pmatrix} 20 \\ -13 \\ 5 \end{pmatrix}$ Calculating magnitudes: $\mathbf{b} \times \mathbf{F}_2 = \sqrt{594}$ $\mathbf{a} \times \mathbf{F}_1 = \sqrt{595}$ so $\mathbf{F}_1$ exerts the torque of greater magnitude	M1 (AO 3.4) A1 (AO 1.1) A1 (AO 1.1) [3]	soi BC BC	
		c	$(\mathbf{a} + \lambda \mathbf{p}) \times \mathbf{F}_1 = \mathbf{a} \times \mathbf{F}_1 + \lambda \mathbf{p} \times \mathbf{F}_1$ but $\mathbf{p}$ and $\mathbf{F}_1$ are parallel implies $\mathbf{p} \times \mathbf{F}_1 = \mathbf{0}$ So $(\mathbf{a} + \lambda \mathbf{p}) \times \mathbf{F}_1 = \mathbf{a} \times \mathbf{F}_1$ as in part (a)	M1 (AO 3.1a) B1 (AO 2.2a) A1 (AO 1.1) [3]	Use distributive property use parallel vectors property in the correct direction conclude	
		d	$\mathbf{F}_3$ exerts no torque means that $\mathbf{a} \times \mathbf{F}_3 = \mathbf{0}$ Vector product = 0 implies that $\mathbf{a}$ and $\mathbf{F}_3$ are parallel, so $\mathbf{F}_3$ acts in the same direction as the line OA	M1 (AO 3.4) E1 (AO 2.2a) [3]	Must not just verify use parallel vectors property in the correct direction	
			Total	10		



Question			Answer/Indicative content	Marks	Part marks and guidance	
30		a	(i) Since $a \neq 0$ and $c \neq 0$, $ac \neq 0 \Rightarrow (ac, b + ad) \in L$ so L is closed under $*$ [1] (ii) $[(a, b) \diamond (c, d)] \diamond (e, f)$ $= (ac, b + ad) \diamond (e, f)$ $= (ace, b + ad + acf)$ $(a, b) \diamond [(c, d) \diamond (e, f)]$ $= (a, b) \diamond (ce, d + cf)$ $= (ace, b + a(d + cf))$ Since $a(d + cf) = ad + acf$ the two are equal, we have associativity (iii) $(a, b) \diamond (c, d) = (ac, b + ad) = (a, b)$ e.g. $\Rightarrow c = 1, d = 0$ i.e. $(1, 0)$ is the identity (iv) $(a, b) \diamond (c, d) = (ac, b + ad) = (1, 0)$ give $c = \frac{1}{a}$ and $d = -\frac{b}{a}$ so $(a, b)^{-1} = \left(\frac{1}{a}, -\frac{b}{a}\right)$ which exists and is in L since $a \neq 0$ an $\frac{1}{a} \neq 0$ d	B1 (AO 2.4) [1] B1 (AO 2.4) M1 (AO 1.1a) A1 (AO 1.1) M1 (AO 2.1) A1 (AO 2.4) [4] M1 (AO 1.1a) A1 (AO 1.1) [3] M1 (AO 3.1a) A1 (AO 2.2) E1 (AO 2.4) [2]	Attempt at $(x \diamond y) \diamond z$ or $x \diamond (y \diamond z)$ for elements x, y, z The other attempted Must state that the two are equal Setting product equal to either component Must note here that the inverse is in L	

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Question			Answer/Indicative content	Marks	Part marks and guidance	
		b	$(a, b) * (a, b) = (1, 0)$ i.e. $(a^2, b(1+a)) = (1, 0)$ For this not to be the identity, $a = -1$ and b arbitrary So the set $\{(1, 0), (-1, b)\}$ for any real b will form a subgroup of order 2	M1 (AO 3.1a) A1 (AO 2.2a) [2]	Looking for any self-inverse element Must choose a value for b or note that any will do	
			Total	12		
31		a	$10^2 = 100 = 2 \times 47 + 6$ so $10^2 \equiv 6 \pmod{47}$	M1 (AO 1.1) A1 (AO 2.4) [2]	Attempted division of 100 by 47 AG Result shown with justification	
		b	$6 \times 8 = 48 = 47 + 1 \equiv 1 \pmod{47}$ so $r = 8$ is the required reciprocal of 6 modulo 47	E1 (AO 2.1) B1 (AO 2.2a) [2]	Correct result clearly demonstrated Correct reciprocal identified, possibly BC	
		c	$10^{46} \equiv 1 \pmod{47}$ since 47 is prime and $\text{hcf}(10, 47) = 1$ The order of 10 mod 47 is then a factor of 46 and since $10^{46} - 1 = (10^{23} - 1)(10^{23} + 1)$, $10^{23} \equiv \pm 1$ and $n = 23$ The only possible smaller divisor of 46 is 2, but (i) showed that 10^2 invalid	B1 (AO 3.1a) E1 (AO 2.4) M1 (AO 2.1) A1 (AO 3.2a) B1 (AO 2.3) [5]	Answer must be justified, not just claimed.	
			Total	9		



Question			Answer/Indicative content	Marks	Part marks and guidance	
32		a	(i) $u_2 = 13, u_3 = 53, u_4 = 213$ (ii) (Auxiliary Equation is $m^2 - 4 = 0$) so that Complementary Solution (CS) is $u_n = A(n^4)$ Particular Solution (PS) is $u_n = a$ with $a - 4a = 1$ i.e. PS is $u_n = -\frac{1}{3}$ (General Solution is $u_n = A(4^n) - \frac{1}{3}$ Using $n = 1, u_1 = 3$, to find the value of A $u_n = \frac{5}{6}(4^n) - \frac{1}{3}$ or equivalent	B1 (AO 1.1) [1] B1 (AO 1.2) B1 (AO 1.1) B1FT (AO 1.1) M1 (AO 1.1a) A1 (AO 1.1) [5]	All three FT provided CS has 1 arbitrary constant, PS none so or $A = \frac{5}{6}$	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		b	(i) Result is true for $n = 1$ since $u_1 = 10 \times 0 + 3 = 3$ Assuming $u_k = 10m + 3$ Then $u_{k+1} = 4(10m + 3) + 1$ $= 10(4m + 1) + 3$ $\Rightarrow$ result true for $n = k + 1$ when true for $n = k$; since true for $n = 1$, it follows that it is true for all positive integers n by induction (ii) E.g. The quadratic residues mod 10 are 0, 1, 4, 9, 5, 6 All u_n's are $\equiv 3 \pmod{10}$ and hence not squares	B1 (AO 2.2a) M1 (AO 2.1) M1 (AO 1.1a) A1 (AO 2.5) E1 (AO 2.4) [5] M1 (AO 3.1a) A1 (AO 3.2a) [2]	Use of the recurrence relation to get the next term Shown to be of required form Suitable 'round-up' of the induction logic Or listing units digits of squares in base 10 May note quadratic non-residues 2, 3, 7 and 8	
			Total	13		



Question			Answer/Indicative content	Marks	Part marks and guidance	
33		a	$\frac{\partial z}{\partial x} = \tan y$ and $\frac{\partial z}{\partial y} = x \sec^2 y$	B1 (AO 1.1) B1 (AO 1.1) [2]		
		b	$x = 1, y = \frac{1}{4}\pi$ substituted into the two partial derivatives $\frac{\partial z}{\partial x} = 1$ and $\frac{\partial z}{\partial y} = 2$ $z = 1$ Equation of tangent plane is $z - 1 = 1(x - 1) + 2(y - \frac{1}{4}\pi)$ i.e. $x + 2y - z = \frac{1}{2}\pi$ oe	M1 (AO 1.1a) A1 (AO 1.1) B1 (AO 1.1) M1FT (AO 1.1a) A1 (AO 1.1) [5]	Both FT three previous values	
			Total	7		



Question			Answer/Indicative content	Marks	Part marks and guidance	
34		a	Aux. Eqn. is $\lambda^2 - 10\lambda + 1 = 0 \Rightarrow \lambda = 5 \pm \sqrt{24} \text{ or } 5 \pm 2\sqrt{6}$ Gen. Soln. is then $u_n = A(5 + 2\sqrt{6})^n + B(5 - 2\sqrt{6})^n$	M1 (AO1.1a) A1 (AO1.1) B1 (AO1.1) [3]	FT their two values of λ	
		b	(i) $\{a_n\} = \{1, 5, \underline{49}, \underline{485}, \dots\}$ and $\{b_n\} = \{0, 2, \underline{20}, \underline{198}, \dots\}$ (ii) $a_0 = 1, a_1 = 5 \Rightarrow 1 = A + B$ and $5 = 5(A + B) + (A - B)2\sqrt{6}$ $\Rightarrow A = B = \frac{1}{2} \text{ i.e.}$ $a_n = \frac{1}{2}(5 + 2\sqrt{6})^n + \frac{1}{2}(5 - 2\sqrt{6})^n$ $b_0 = 0, b_1 = 2 \Rightarrow 0 = A + B$ and $2 = 5(A + B) + (A - B)2\sqrt{6}$ $\Rightarrow A = -B = \frac{1}{2\sqrt{6}} \text{ i.e.}$ $b_n = \frac{1}{2\sqrt{6}}(5 + 2\sqrt{6})^n - \frac{1}{2\sqrt{6}}(5 - 2\sqrt{6})^n$ Either Calling $\alpha = 5 + 2\sqrt{6}$ and $\beta = 5 - 2\sqrt{6}$ and substituting both in $(a_n)^2 - 6(b_n)^2 = \frac{1}{4}\{\alpha^{2n} + 2(\alpha\beta)^n + \beta^{2n}\} - 6 \times \frac{1}{24}\{\alpha^{2n} - 2(\alpha\beta)^n + \beta^{2n}\}$ $= (\alpha\beta)^n = 1$, by the	B1 (AO1.1) [1] M1 (AO1.1a) A1 (AO1.1) M1 (AO1.1a) A1 (AO1.1) M1 (AO3.1a) A1 (AO2.1) A1 (AO2.5) A1 (AO2.2a)	All highlighted terms required Sim. eqns. may be solved by GC, for instance Sim. eqns. may be solved by GC, for instance The shorthand is helpful but not necessary	



Question			Answer/Indicative content	Marks	Part marks and guidance	
			"difference of two squares" (or product of roots) $O (a_n)^2 - 6(b_n)^2 = (a_n - \sqrt{6}b_n)(a_n + \sqrt{6}b_n)$ $r = \frac{1}{2} \{ \alpha^n + \beta^n - \alpha^n + \beta^n \} \cdot \frac{1}{2} \{ \alpha^n + \beta^n + \alpha^n - \beta^n \}$ $= (\alpha\beta)^n = 1$ as above Partial Alternative using Induction: Result true for $n = 0, 1, 2, 3$ Assuming $(a_{n-1})^2 - 6(b_{n-1})^2 = 1$ and $(a_n)^2 - 6(b_n)^2 = 1$ $(a_{n+1})^2 - 6(b_{n+1})^2 = (10a_n - a_{n+1})^2 - 6(10b_n - b_{n+1})^2$ $= 100(a_n^2 - 6b_n^2) + (a_{n+1}^2 - 6b_{n+1}^2) - 20(a_n a_{n+1} - 6b_n b_{n+1})$ $= 101 - 20(a_n a_{n+1} - 6b_n b_{n+1})$ The bracket can be addressed as above, or by an inductive proof in itself. (ii) Either i) For large n, $(a_n)^2 \approx 6(b_n)^2$ $\Rightarrow \frac{a_n}{b_n} \rightarrow \sqrt{6}$ Or $\beta < 1 \Rightarrow a_n \rightarrow \frac{1}{2} \alpha^n$ and $b_n \rightarrow \frac{1}{2\sqrt{6}} \alpha^n$ so that $\frac{a_n}{b_n} \rightarrow \frac{\frac{1}{2} \alpha^n}{\frac{1}{2\sqrt{6}} \alpha^n} = \sqrt{6}$	M1 M1 A1 A1 B1 M1 M1 M1 A1 [8] M1 (AO2.1) A1 (AO2.1) M1 A1 [2]	$\alpha\beta = 1$ must at least be stated Baseline case(s) ... ≥ 2 (consecutive) must be noted Assuming <i>two</i> previous results Attempt at $(n+1)$th case via the r.r. Appropriate pairing of terms	



Question			Answer/Indicative content	Marks	Part marks and guidance
			Total	14	



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Question			Answer/Indicative content	Marks	Part marks and guidance	
35		a	If $f^2(x) = x$ or $f^3(x) = x$ then it will generate a group of order 2 or 3.	B1(A02.3) B1 (A02.2a) [2]		
		b	Since $\det(\text{product}) = \text{product of dets}$ we need $\det(M^6) = (\det M)^6 = \det(I) = 1$ $\Rightarrow \det(M) = \pm 1$	E1 (AO 3.1a) E1 (AO 2.1) [2]	Enter text here.	
		c	Want $\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} \cos \frac{2\pi}{6} & -\sin \frac{2\pi}{6} \\ \sin \frac{2\pi}{6} & \cos \frac{2\pi}{6} \end{pmatrix}$	B1 (AO 3.4) [1]	i.e. a generator of the group of rotations through $\frac{\pi}{3}$ multiples of	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		d	(i) $\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$ gives $f(x) = \frac{\frac{1}{2}x - \frac{\sqrt{3}}{2}}{\frac{\sqrt{3}}{2}x + \frac{1}{2}} \text{ or } \frac{x - \sqrt{3}}{x\sqrt{3} + 1}$ [1] (ii) $M^3 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$ gives $f^3(x) = \frac{-x+0}{0x-1} = x, \text{ so } f \text{ not of order } 6$ [2] (iii) However, taking $\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} \cos \frac{\pi}{6} & -\sin \frac{\pi}{6} \\ \sin \frac{\pi}{6} & \cos \frac{\pi}{6} \end{pmatrix}$ M1(AO3.5c) A1 gives $f(x) = \frac{x\sqrt{3}-1}{x+\sqrt{3}}$ for which $f^2(x) = \frac{x-\sqrt{3}}{x\sqrt{3}+1} \text{ and}$ $f^3(x) = \frac{-1}{x} \Rightarrow f^6(x) = x$ Must state, or have shown, that both $f^2(x)$ and $f^3(x) \neq x$ A1 (AO2.4) [4]	B1 (AO 3.5a) B1 (AO 3.5b) B1 (AO 3.1a) M1(AO3.5c) A1 M1(AO3.2a) A1 (AO2.4)	* BC Must state the conclusion and not stop with the working Another rotation matrix A correct suggestion Attempt at powers of f up to f^6	
			Total	12		



Question			Answer/Indicative content	Marks	Part marks and guidance	
36		a	$N = \binom{p-1}{r} = \frac{(p-1)!}{r!(p-r-1)!}$ so $r! N = (p-1)(p-2) \dots (p-r)$ $\equiv (-1)(-2) \dots (-r) \pmod{p}$ $= (-1)^r (r!)$ Since all terms in $r!$ are $< p$, $\text{hcf}(r!, p) = 1$ so we can divide both sides by $r!$ to obtain $N \equiv (-1)^r \pmod{p}$	M1(A01.1a) A1(A01.1) M1(A02.1) A1(A02.5) A1(A02.4) [5]	Use of correct factorial form for N Correct with all fractions removed AG obtained with full justification	
		b	$M = \binom{2p}{p} = \frac{(2p)!}{p! p!} \Rightarrow p! M = \frac{(2p)!}{(2p-1)(2p-2) \dots (2p-(p-1))}$ $\Rightarrow (p-1)! M = 2(2p-1)(2p-2) \dots (2p-(p-1))$ $\Rightarrow (p-1)! M \equiv 2(-1)(-2) \dots (-(p-1)) \pmod{p}$ $= 2(-1)^{p-1} (p-1)!$ Since all terms in $(p-1)!$ are $< p$, $\text{hcf}((p-1)!, p) = 1$ we can divide both sides by $(p-1)!$; also $(p-1)$ is even so $(-1)^p = 1$. Hence $M \equiv 2 \pmod{p}$	M1(A03.1a) A1(A01.1) M1(A03.1a) M1(A02.1) A1(A01.1) A1(A02.2a) [6]	Use of correct factorial form for M Correct with all fractions removed AG obtained with full justification	
			Total	11		



Question			Answer/Indicative content	Marks	Part marks and guidance	
37		a	(i) Use of volume formula $V = \frac{1}{6}x \cdot y \times z$ in any valid setting Genuine attempt at a suitable scalar triple product $(a \cdot b \times c)$ for $OABC$ $\Rightarrow V_{OABC} = \frac{1}{6}abc$ Second scalar triple product attempted for $PABC$ $(p - a) \cdot (p - b) \times (p - c) = \begin{pmatrix} 0 \\ b \\ c \end{pmatrix} \cdot \begin{pmatrix} a \\ 0 \\ c \end{pmatrix} \times \begin{pmatrix} a \\ b \\ 0 \end{pmatrix} = (2abc)$ $\Rightarrow V_{PABC} = \frac{1}{3}abc$ (ii Since $V(OABC) :$ $V(PABC) = 1 : 2$ and the area of the base-plane ABC is common to both tetrahedra, the “heights” are also in the ratio $1 : 2$	M1 (A01.1a) M1 (A01.1) A1(A01.1) M1(A01.1) A1(A01.1) [5] B1(A03.1a) B1(A02.4) [2]	Any arrangement of a, b, c By the determinant method or long-hand Must have appropriate subtracted components	

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Question			Answer/Indicative content	Marks	Part marks and guidance	
		b	(i) $\mathbf{n} = (\mathbf{b} - \mathbf{a}) \times (\mathbf{c} - \mathbf{a})$ (e.g.) $= \begin{pmatrix} -a \\ b \\ 0 \end{pmatrix} \times \begin{pmatrix} -a \\ 0 \\ c \end{pmatrix} = \begin{pmatrix} bc \\ ca \\ ab \end{pmatrix}$ (ii) Area $\Delta ABC = \frac{1}{2} \mathbf{n}$ where $\mathbf{n}$ = vector answer to (a) Then $\frac{1}{6} abc = \frac{1}{3}$ (area ΔABC) (dist. O to Π) $\Rightarrow$ dist. P to $\Pi = 2 \times$ dist. O to Π $= \frac{2abc}{\sqrt{a^2b^2 + b^2c^2 + c^2a^2}}$ Alternative Plane ABC has eqn. $\mathbf{r} \cdot \mathbf{n} = \mathbf{a} \cdot \mathbf{n} = abc$ (e.g.) using $\mathbf{n}$ from (a) Dist. O to $\Pi =$ this RHS $\div \text{their } \mathbf{n}$ $\Rightarrow$ dist. P to $\Pi = 2 \times$ dist. O to Π $= \frac{2abc}{\sqrt{a^2b^2 + b^2c^2 + c^2a^2}}$	M1(A01.1a) A1(A01.1) [2] B1(A03.1a) M1(A01.1a) A1(A02.2a) B1 M1 A1 [3]	Not required (yet) in terms of a, b, c Use of formula	
			Total	12		



Question			Answer/Indicative content	Marks	Part marks and guidance	
38		a	(i) $f_x = 2xyz + 2y^2z + 3yz^2 - 24yz$ or $yz(2x + 2y + 3z - 24)$ * $f_y = x^2z + 4xyz + 3xz^2 - 24xz$ or $xz(x + 4y + 3z - 24)$ $f_z = x^2y + 2xy^2 + 6xyz - 24xy$ or $xy(x + 2y + 6z - 24)$ (ii Setting all p.d.s to zero) $2x + 2y + 3z = 24$ All three are zero ($x, y, z \neq 0$) when $x + 4y + 3z = 24$ $x + 2y + 6z = 24$ $x = 6, y = 3, z = 2$ $w = -216$	M1(A01.1a) A1(A01.2) A1(A01.1) A1(A01.1) [4] M1(A01.1a) M1(A02.1) M1(A01.1a) A1(A01.1) A1(A01.1) [5]	Good attempt at (at least) one partial derivative Factoring out the non-zero terms and setting up a solvable system of equations Solving a 3×3 system of equations (BC, for instance) Correct cao (no need to mention these are a, b, c, d)	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		b	Want (e.g.) x large and $-ve$, y small and $-ve$, z small and $+ve$ e.g. $x = -10$, $y = -1$, $z = 1$ so that $w = -330 < -216$	M1(A03. 1a) A1(A03. 2a) [2]	Correct idea Demonstrated correctly	
			Total	11		



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Question			Answer/Indicative content	Marks	Part marks and guidance	
39		a	$\frac{dx}{dt} = 3t^2 - 3 \text{ and } \frac{dy}{dt} = 6t$ $\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = 9t^4 - 18t^2 + 9 + 36t^2$ $L = \int_0^5 (3t^2 + 3) dt$ $= \left[t^3 + 3t \right]_0^5$ $= 140$	B1 (A01.1) M1(A01.1) A1(A02.1) M1(A01.1a) A1(A02.1) A1(A01.1) [6]	DR Both correct Attempted Must be in the form of a perfect square (here or later) Correct arc-length formula statement (FT the integrand) DR Correct integration of (at least) a quadratic; ignore limits	
		b	$A = 2\pi \int_0^5 (3t^2 - 1)(3t^2 + 3) dt$ $= 6\pi \int_0^5 (3t^4 + 2t^2 - 1) dt$ $= 6\pi \left[\frac{3}{5}t^5 + \frac{2}{3}t^3 - t \right]$ $= 11720\pi$	M1(A01.1a) A1(A01.1) A1(A02.1) A1(A01.1) [4]	DR Correct surface area formula statement (FT the integrand) Correct, in an integrable form Correct integration of (at least) a quartic; ignore limits	
			Total	10		

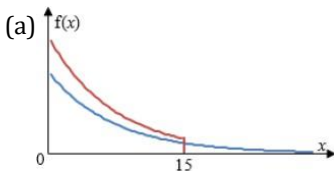


Question			Answer/Indicative content	Marks	Part marks and guidance	
40		a	$100011_n = n^5 + n + 1$	B1(A01.1) [1]		
		b	$n^5 + n + 1 = (n^2 + n + 1)(n^3 - n^2 + 1)$	M1(A01.1) A1(A02.1) [2]	Attempt at long division or equivalent Correct cubic factor	
		c	For $n \geq 2$, $n^2 + n + 1 \geq 7$ and $n^3 - n^2 + 1 \geq 5$ and both factors $> 1 \Rightarrow N$ composite	B1(A02.2a) E1(A02.4) [2]	Statement that both factors > 1 $\Rightarrow N$ composite Supporting working (any convincing justification)	
			Total	5		



Question			Answer/Indicative content	Marks	Part marks and guidance	
41		a	DR $\int_0^{\infty} x\lambda e^{-\lambda x} dx$ $= \left[-xe^{-\lambda x} \right]_0^{\infty} - \int_0^{\infty} -e^{-\lambda x} dx$ $= \lim_{k \rightarrow \infty} \left(\left[-xe^{-\lambda x} - \frac{1}{\lambda} e^{-\lambda x} \right]_0^k \right)$ $= \frac{1}{\lambda} \text{ so } \lambda = \frac{1}{5.0} = 0.2$	M1 (AO 1.1a) M1 (AO 1.1) E1 (AO 2.4) A1 (AO 1.1) [4]	$\int_{20}^{\infty} 0.2e^{-0.2x} dx$ Use Parts used explicitly, $e^{-\lambda}$ integrated or equivalent consideration of limits Correctly obtain $\frac{1}{\lambda}$ and $\lambda = 0.2$, www accept $= 0 - \left[\frac{1}{0.2} e^{-0.2x} \right]_0^{\infty}$	
		b	$\int_{20}^{\infty} 0.2e^{-0.2x} dx$ $= 0.0183156 \dots$	M1 (AO 3.1b) A1 (AO 3.4) [2]	Or $1 - \int_0^{20} 0.2e^{-0.2x} dx$ BC Awrt 0.0183	



Question			Answer/Indicative content	Marks	Part marks and guidance		
		c	(a)  (b)	B1 (AO 1.1) [1] M1 (AO 3.5c) A1 (AO 3.3) [2]	Correct shape for $y = \lambda e^{-\lambda x}$, asymptotic to x-axis only Any curve truncated at 15 More than half above their curve from part (i)	Don't need vertical line i.e., awareness that total area is the same	
			Total	9			



Question			Answer/Indicative content	Marks	Part marks and guidance	
42		a	$X = \frac{1}{\sqrt{3}} \begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix} \Rightarrow X^2 = \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}, X^3 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & -2 \\ 2 & -1 \end{pmatrix},$ $X^4 = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, X^5 = \frac{1}{\sqrt{3}} \begin{pmatrix} -1 & -1 \\ 1 & -2 \end{pmatrix}, X^6 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$ Note that $X^7 = -X$ etc. so that none of X^7 to X^{11} is equal to I $\Rightarrow X^{12} = I$ and so $n(M) = 12$ because the powers of X now cycle every 12th power but no sooner	M1 (AO 1.1a) A1 (AO 1.1) A1 (AO 2.2a) E1 (AO 2.4) [4]	Attempt at powers of X (possibly BC) Correct up to X^6 (NB – not all needed) Candidates may calculate each of these BC Correctly deduced Clear statement needed to justify $n(M) = 12$	
		b	The set is closed (each element is one of the 12 powers of X); matrix multiplication is known to be associative; the multiplicative identity I is in M; and the inverse of X^k is X^{12-k} for all suitable k ... so we have a group G The group is cyclic since it is generated by X (or X^5 or X^7 or X^{11}) oe	B1 (AO 2.1) B1 (AO 1.1) B1 (AO 1.1a) B1 (AO 1.1) E1 (AO 2.4) [5]	One for each group axiom Inverses may be explicitly paired-up Only one generator needs to be mentioned here	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		c	Proper subgroups of G are $H_2 = \{X^6, I\}$, $H^3 = \{X^4, X^8, I\}$, $H_4 = \{X^3, X^6, X^9, I\}$, $H_4 = \{X^2, X^4, X^6, X^8, X^{10}, I\}$ The above subgroups are those generated as follows: $H_2 = (X^6)$, $H_3 = (X^4) = (X^8)$, $H_4 = (X^3) = (X^9)$ and $H_6 = (X^2) = (X^{10})$ and are thus unique Mention that <i>Lagrange's Theorem</i> states that $H \mid G$ so there are no subgroups of orders 5, 7, 8, 9, 10 or 11 OR For listing the subgroups (as above) The elements have orders $\{12, 6, 4, 3, 12, 2, 12, 3, 4, 6, 12, 1\}$ and one can argue from the orders of elements that (e.g.) : a subgroup of order 2 must contain I and an element of order 2 $\Rightarrow H^2$ uniquely; a subgroup of order 3 must contain I and an element of order 3 $\Rightarrow H^3$ uniquely, since there are only two such elements and they are an inverse-pair; similarly for H^6 Then as above for no subgroups of orders 5, 7, 8, 9, 10 or 11	B1 (AO 3.1a) B1 (AO 1.1) B1 (AO 3.1a) E1 (AO 3.1a) B1 (AO 2.2a) [5] B1 B1 B1 E1 B1	Any two correct; All four correct (+ no extras – ignore inclusion of $\{I\}$ and/or G) Condone omission to mention that any subgroup containing X^5 or X^7 or X^{11} is G (since they are generators)	
			Total	14		



Question			Answer/Indicative content	Marks	Part marks and guidance	
43		a	$I_1 = \int_0^{\sqrt{3}} t\sqrt{1+t^2} dt = \left[\frac{1}{3}(1+t^2)^{\frac{3}{2}} \right]_0^{\sqrt{3}}$ $= \frac{1}{3}(8-1) = \frac{7}{3}$	M1 (AO 1.1a) A1 (AO 1.1) [2]	By "recognition" (reverse <i>Chain Rule</i>) integration or any other suitable method (e.g. by substitution): MUST have $k(1+t^2)^{1.5}$ AG from correct working	
		b	$I_n = \int_0^{\sqrt{3}} t^n \sqrt{1+t^2} dt = \int_0^{\sqrt{3}} t^{n-1} \cdot t\sqrt{1+t^2} dt$ $= \left[t^{n-1} \cdot \frac{1}{3}(1+t^2)^{\frac{3}{2}} \right]_0^{\sqrt{3}} - \int_0^{\sqrt{3}} (n-1)t^{n-2} \cdot \frac{1}{3}(1+t^2)^{\frac{3}{2}} dt$ $\Rightarrow 3I_n = (\sqrt{3})^{n-1} \cdot 8 - 0 - (n-1) \int_0^{\sqrt{3}} t^{n-2} (1+t^2)^{\frac{3}{2}} dt$ $\Rightarrow 3I_n = 8(\sqrt{3})^{n-1} - (n-1)\{I_{n-2} + I_n\}$ $\Rightarrow (n+2)I_n = 8(\sqrt{3})^{n-1} - (n-1)I_{n-2}$	M1 (AO 2.1) A1 A1 (AO 1.1, 1.1) M1 (AO 3.1a) A1 (AO1.1) A1 (AO 2.2a) [6]	Correct splitting and attempt at <i>integration by parts</i> One for each correct part Splitting up the power of $(1+t^2)$ suitably 2nd integral correctly identified in terms of I's AG Legitimately obtained from correct rearrangement of I terms	
		c	$\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = (30t^2)^2 + (30t)^2$ $= 900t^2(1+t^2) \text{ so } I$ $A = 2\pi \int_0^{\sqrt{3}} 15t^2 \cdot 30t\sqrt{1+t^2} dt$ $= 900\pi I_3 \text{ i.e. } k = 900, m = 3$ Use of $n = 3$ in Reduction $5 I_3 = 8(\sqrt{3})^5 - 2 \times \frac{7}{3} \Rightarrow I_3 = \frac{38}{15}$ Formula: so that $A = 3480\pi$	M1 (AO 1.1a) A1 (AO 1.1) M1 (AO 2.1) A1 (AO 1.1) M1 (AO 3.1a) A1 (AO 2.2a) [6]	Attempted use of formula Correct, any form Formula used correctly with relevant terms in t cao	



Question			Answer/Indicative content	Marks	Part marks and guidance
			Total	14	



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Question			Answer/Indicative content	Marks	Part marks and guidance	
44		a	Characteristic Equation is $\lambda^2 - 1.3\lambda - 0.3 = 0$ $\Rightarrow \lambda = 1.5, -0.2$ $\Rightarrow$ General Solution is $X^n = A \times 1.5^n + B \times (-0.2)^n$ Use of $X_0 = 12$ and $X_1 = 1$ to obtain equations in A, B: i.e. $12 = A + B$ & $1 = 1.5A - 0.2B$ Solving $\Rightarrow A = 2, B = 10$ Solution is $X_n = 2 \times 1.5^n + 10 \times (-0.2)^n$ For large n, $(-0.2)^n \rightarrow 0$ so $X_n \rightarrow 2 \times 1.5^n$ which is of the form $a r^n$, hence a GP	M1 (AO 1.1a) A1 (AO 1.1) B1 (AO 1.2) M1 (AO 1.1a) M1 (AO 1.1) A1 (AO 2.5) [6] B1 (AO 3.1b) E1 (AO 2.2a) [2]	BC FT their λs Eqns. solved simultaneously (e.g. BC) cao brackets required OR $3 \times 1.5^{n-1}$, of the form $a r^{n-1}$, hence a GP [MUST have some explanation that this is a GP]	

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Question			Answer/Indicative content	Marks	Part marks and guidance	
		b	Tabulating the given sequence (or using calculator equation solver) $X_{32} \approx 862880 < 1\,000\,000$ and $X_{33} \approx 1294320 > 1\,000\,000$ so $n = 33$ $X_n = 2 \times 15^n > 1\,000\,000$ $\Rightarrow n > \frac{\log 500000}{\log 1.5} = 32.36... \text{ so } n = 33$	M1 (AO 3.1a) A1 (AO 3.2a) [2] M1 (AO 2.1) A1 (AO 1.1) [2]	BC or manual calculation Properly justified Attempt at solving (logs not essential) Properly justified	
		c	$X_{n+2} = \text{INT}[1.3 X_{n+1} + 0.3 X_n]$ $X_{n+2} = \text{INT}[1.3 X_{n+1} + 0.3 X_n + 1]$ or $X_{n+2} = \text{INT}[1.3 X_{n+1} + 0.3 X_n] + 1$ If $1.3 X_{n+1} + 0.3 X_n$ were <i>exactly</i> an integer value at any stage, then the next X_n would be too large by 1	B1 (AO 3.3) [1] B1 (AO 3.5c) [1] B1 (AO 3.5b) [1]		
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Question			Answer/Indicative content	Marks	Part marks and guidance	
45		a	1, 4, 9, 5, 3 The 5th-powers (mod 11) are 0, 1, 10 so $LHS \equiv 0, 1, 10 \pmod{11}$ $2017 \equiv 4 \pmod{11}$ so $RHS \equiv 4, 5, 8, 2, 9, 7 \pmod{11}$ Since LHS and RHS are not equal modulo 11, the equation has no solution in integers	B1 (AO 1.1a) B1 (AO 1.1) [2] M1 (AO 3.1a) A1 (AO 1.1) M1 (AO 1.1a) A1 (AO 2.2a) E1 (AO 2.4) [5]	Give 1st B1 for any 3 correct 2nd B1 for all 5 and allow 0, no incorrect extras condone (0,)1, 4, 9, 5, 3, 3, 5, 9, 4, 1 Looking at 5th powers mod 11 All 3 found with no extras $2017 \equiv 4 \pmod{11}$ BC FT each of (a)'s residues + 4 Must follow from fully correct working	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		b	DR $\text{mod } 9, M \equiv 2^{12} - 1 = (2^3)^4 - 1 \equiv (-1)^4 - 1 = 1 - 1 = 0$ OR $11^{12} - 1 = (11^6 - 1)(11^6 + 1) = (11^3 - 1)(11^3 + 1)(11^6 + 1)$ and $11^3 + 1 = 1332$, which is divisible by 9 using the standard divisibility test $\text{mod } 5, M \equiv 1^{12} - 1 = 0$ or via above factorisation, noting that $5 \mid 11^3 - 1 = 1330$ The factor $11^6 - 1 \equiv 0 \pmod{7}$ by <i>FLT</i> since $\text{hcf}(11, 7) = 1$ and 7 is prime $M = 11^{12} - 1 \equiv 0 \pmod{13}$ by <i>FLT</i> since $\text{hcf}(11, 13) = 1$ and 13 is prime $\text{mod } 61, 11^2 = 121 \equiv -1 \Rightarrow 11^{12} - 1 \Rightarrow (-1)^6 - 1 = 1 - 1 = 0$ NB $M = 11^{12} - 1 = 3 \ 138 \ 428 \ 376 \ 720 = 2^4 \cdot 3^2 \cdot 5 \cdot 7 \cdot 13 \cdot 19 \cdot 37 \cdot 61 \cdot 11 \cdot 17$ (Also, M has $11^3 + 1$ as a factor, and $11^3 + 1 = (11 + 1)(11^2 - 11 \cdot 1 + 1^2)$ by the <i>sum-of-two-cubes</i> factorisation, so it follows that M is divisible by $11^2 - 11 \cdot 1 + 1^2 = 111 = 3 \times 37$; that is, M is a multiple of 37)	B1 (AO 3.1a) B1 (AO 3.1a) B1 (AO 2.1) B1 (AO 2.4) B1 (AO 2.2a) [5]	Using the <i>difference-of-two-squares factorisation</i> (e.g.) NB: can argue via $10 \mid (11^n - 1) \forall$ integers $n \in \mathbb{N}$ by the <i>Factor Theorem</i>; etc. Any reasonable, clearly explained method is acceptable NB <i>FLT</i> is <i>Fermat's Little Theorem</i>	
			Total	12		



Question			Answer/Indicative content	Marks	Part marks and guidance	
46		a	$\frac{\partial z}{\partial x} = \frac{1}{y} \sin y - \frac{y}{x} \sin x - \frac{y}{x^2} \cos x$ $\frac{\partial z}{\partial y} = \frac{x}{y} \cos x - \frac{x}{y^2} \sin y + \frac{1}{x} \cos x$ oe oe	M1 (AO1.1a) A1 (AO1.1) M1 (AO1.1a) A1 (AO1.1) [4]	Partial differentiation w.r.t x (including use of the <i>Product or Quotient Rule</i>) Partial differentiation w.r.t y (including use of the <i>Product or Quotient Rule</i>)	
		b	Wh $x = y = \frac{1}{4}\pi, z = \sqrt{2}$ en $\frac{\partial z}{\partial x} = \frac{-1}{\sqrt{2}} \text{ an } \frac{\partial z}{\partial y} = \frac{1}{\sqrt{2}}$ oe d Eqn. of tangent-plane is $z = \frac{-1}{\sqrt{2}}(x - \frac{1}{4}\pi) + \frac{1}{\sqrt{2}}(y - \frac{1}{4}\pi) + \sqrt{2}$ $\Rightarrow x - y + z\sqrt{2} = 2$ oe	B1 (AO 1.1) B1 (AO 1.1) B1 (AO 1.1) M1 (AO 2.2a) A1 (AO 1.1) [5]		
		c	$\begin{pmatrix} 1 \\ -1 \\ \sqrt{2} \end{pmatrix}$ or any suitable multiple	B1 (AO 2.2a) [1]	FT from their answer to (b)	
			Total	10		



Question			Answer/Indicative content	Marks	Part marks and guidance	
47			$\mathbf{x} = \mathbf{b} - \mathbf{a} = \begin{pmatrix} 2 \\ 2 \\ -9 \end{pmatrix}, \mathbf{y} = \mathbf{c} - \mathbf{a} = \begin{pmatrix} 5 \\ 0 \\ -2 \end{pmatrix}$ $\mathbf{z} = \mathbf{d} - \mathbf{a} = \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}$ g. $\text{Volume} = \frac{1}{6} \mathbf{x} \cdot \mathbf{y} \times \mathbf{z} $ $\begin{vmatrix} 2 & 2 & -9 \\ 5 & 0 & -2 \\ -1 & 1 & 2 \end{vmatrix} = -57$ $\Rightarrow \text{Volume} = 9.5$	M1* (AO 1.1a) A1 (AO 1.1) dep*M1 (AO 1.1a) B1 (AO 1.1) A1 (AO 2.2a) [5]	Subtraction to find any 3 sides of the tetrahedron Attempted use of formula A correct (non-zero) scalar triple product (possibly BC). Condone sign error FT det; must be positive final answer	
			Total	5		



Question	Answer/Indicative content	Marks	Part marks and guidance
48	$x \equiv 4 \equiv 11, 18, 25, \dots \pmod{7}$ $\Rightarrow x \equiv 25 \pmod{7}$ So $x \equiv 25 \pmod{7}$ and $x \equiv 25 \pmod{41}$ $\Rightarrow x \equiv 25 \pmod{287}$ since $\text{hcf}(7, 41) = 1$ OR $x = 7a + 4 = 41b + 25$ $\Rightarrow 7(a - 3) = 41b \Rightarrow b = 7c$ since $\text{hcf}(7, 41) = 1$ $\Rightarrow a = 41c + 3 \Rightarrow x = 7(41c + 3) + 4$ i.e. $x = 287c + 25 \ (c \in \mathbb{Z})$ OR Reciprocal of $41 \equiv 6 \pmod{7}$ is 6 and Reciprocal of $7 \pmod{41}$ is 6 The <i>Chinese Remainder Theorem</i> guarantees a unique solution mod 7×41 since $\text{hcf}(7, 41) = 1$ Solution is $x = 287n + r, n \in \mathbb{Z}$, where $r = 4 \times 6 \times \frac{287}{7} + 25 \times 6 \times \frac{287}{41}$ $\pmod{287}$ i.e. $r = 2034 \pmod{287} \equiv 25$	M1 (AO 1.1a) A1 (AO 1.1) M1 (AO 2.1) A1 (AO 2.2a) E1 (AO 2.4) [5] M1 M1 E1 M1 A1 M1 A1 E1 M1 A1	Considering equivalents to $4 \pmod{7}$ Putting the two together OR $x = 287n + 25, n \in \mathbb{Z}$ Eliminating either a or b Obtaining x in terms of a single parameter Full implementation of the CRT



Question			Answer/Indicative content	Marks	Part marks and guidance	
			Total	5		



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Question			Answer/Indicative content	Marks	Part marks and guidance	
49		i	$u = x^n \Rightarrow du = nx^{n-1} dx$ $dv = \sqrt{1-x} dx \Rightarrow v = -\frac{2}{3}(1-x)^{3/2}$ $\Rightarrow I_n = \left[-x^n \frac{2}{3}(1-x)^{3/2} \right]_0^1 + \frac{2}{3}n \int_0^1 x^{n-1}(1-x)^{3/2} dx$ $\Rightarrow I_n = 0 + \frac{2}{3}n \int_0^1 x^{n-1}(1-x)\sqrt{1-x} dx$ $= \frac{2}{3}n(I_{n-1} - I_n)$ $\Rightarrow \left(1 + \frac{2}{3}n\right)I_n = \frac{2n}{3}I_{n-1} \Rightarrow I_n = \frac{2n}{2n+3}I_{n-1}$	M1 A1 A1 M1 A1 [5]	Integrate by parts 1 st term must be seen and later made = 0 Second term Dealing with $(1-x)^{3/2} - (1-x)\sqrt{1-x}$ And converting to I_n and I_{n-1} AG	Examiner's Comments In part (i), candidates realised that they needed to use integration by parts. Those who chose initially to define "u" and "dv" the wrong way round soon got themselves into trouble and they were then unable to realise their error, choosing instead to give up the question.
		ii	$\frac{2n}{2n+3} < 1 \Rightarrow I_n < I_{n-1}$	M1 A1 [2]	Examiner's Comments Part(ii) was found difficult by many candidates who did not demonstrate what was required conclusively.	



	Total	10
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Question			Answer/Indicative content	Marks	Part marks and guidance	
50		i	16	B1 [1]		Examiner's Comments This was another question which was mostly answered correctly. Occasional wrong answers of 4, 24 or 256 were seen.
		ii	$\begin{pmatrix} 1 & 0 \\ 0 & b \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & c \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & bc \end{pmatrix} \in H$ so closed $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \in H$ so contains identity $\begin{pmatrix} 1 & 0 \\ 0 & b \end{pmatrix}^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & b^{-1} \end{pmatrix} \in H$ so contains inverses so is (sub) group	B1 B1 B1 [3]	If three items dealt with as in scheme, but fail to say "in H " then deduct one mark. Must conclude to gain all 3 marks. Must conclude and not address commutativity to gain all 3 marks.	Examiner's Comments Most candidates addressed the conditions necessary in a subgroup, but answers were almost always insufficiently clear or precise. In checking for identity, closure and inverses, the commonest deficiency was to fail to say that elements belonged to H , sometimes, instead, saying that they belonged



Question			Answer/Indicative content	Marks	Part marks and guidance	
					to G or “the group”. The best answers addressed: ♦ closure by considering the product of two general elements of H and concluded that the product was still in H, ♦ existence of identity, concluding that it was also in H, ♦ existence of all inverses by computing the inverse of a general element and concluding that it, too, was in H. It should be noted that consideration of associativity is irrelevant for subgroups, this being automatically inherited. It did not, however, draw any penalty, except in the case where the candidate was clearly considering commutativity rather than associativity. A more important consideration (unnecessary here) is whether the set in question is a subset of the parent group. It should also be noted, in consideration of closure, it is inadequate to simply consider the square of a general element.	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$ k $ is a factor of their "16"	M1	Use of Lagrange	
			$ H = 4$ so 4 is a factor of $ K $	M1	or $ k \geq 4$, if 1 st M1	
			so $ K $ 4,8 or 16	A1	awarded	
			proper subgroups so proper factors	A1	May be implied	
			so $ K = 8$	[4]	Complete argument.	
					Examiner's Comments These last two parts challenged even the strongest candidates. Many candidates correctly attempted part (iii) by considering the possible orders of subgroups. However, there was a lack of precision of mathematical argument in many cases. It was crucial to explain explicitly that the order of K is a factor or divisor of $ G (=16)$, and that the order of K is a multiple of $ H (=4)$. To completely justify the solution of "8" it was also necessary to explicitly link the "properness" to the elimination of the other possibilities rather than conflating it with Lagrange. There were many attempts without the above elements merely producing lists of possible orders for K and then whittling them down to 8.	



Question			Answer/Indicative content	Marks	Part marks and guidance		
		iv	Identifies correct subgroup If $\begin{pmatrix} i & 0 \\ 0 & b \end{pmatrix} \in K$ then $\begin{pmatrix} i & 0 \\ 0 & b \end{pmatrix}^2 = \begin{pmatrix} -1 & 0 \\ 0 & b^2 \end{pmatrix} \in K$ If $\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \in K$ for some b then multiplying by elements of H gives $\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$ for all b But this gives more than 8 elements So $\begin{pmatrix} i & 0 \\ 0 & b \end{pmatrix} \notin K$ Similarly $\begin{pmatrix} -i & 0 \\ 0 & b \end{pmatrix} \notin K$ so $K = \left\{ \begin{pmatrix} \pm 1 & 0 \\ 0 & b \end{pmatrix} : b^4 = 1 \right\}$	B1 M1 M1 A1 M1dep A1 [6]	At any stage in solution Possibly in isolation from matrix Dep on both previous M marks being gained For full argument Examiner's Comments This demonstration was treated adequately by only a tiny number of candidates. Many candidates gained one mark for identifying the elements of K, and a possible second mark for making a start on trying to show that K could not contain any matrix of the form $\begin{pmatrix} \pm i & 0 \\ 0 & b \end{pmatrix}$. The attempted arguments here were invariably logically unsound, and could therefore not gain		



Question			Answer/Indicative content	Marks	Part marks and guidance	
					credit in a topic of this sort. The point almost universally missed was that if K contained a particular $\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$ matrix, then it follows by multiplying by a general element of H that K contains $\begin{pmatrix} a & 0 \\ 0 & c \end{pmatrix} \text{ for all } c.$ Consequently, to show that if K contained some $\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$ with $a = \pm i$, it would also contain the square $\begin{pmatrix} -1 & 0 \\ 0 & b^2 \end{pmatrix}$ was to show that it would generate an extra 8 elements in addition to those of H, thereby making the order of K too large.	
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Question			Answer/Indicative content	Marks	Part marks and guidance																										
51		i	<table><tr><td></td><td>1</td><td>5</td><td>7</td><td>11</td></tr><tr><td>1</td><td>1</td><td>5</td><td>7</td><td>11</td></tr><tr><td>5</td><td>5</td><td>1</td><td>11</td><td>7</td></tr><tr><td>7</td><td>7</td><td>11</td><td>1</td><td>5</td></tr><tr><td>11</td><td>11</td><td>7</td><td>5</td><td>1</td></tr></table>		1	5	7	11	1	1	5	7	11	5	5	1	11	7	7	7	11	1	5	11	11	7	5	1	B1 B1 [2]	Twelve entries correct All correct	
	1	5	7	11																											
1	1	5	7	11																											
5	5	1	11	7																											
7	7	11	1	5																											
11	11	7	5	1																											
		ii	3.3 = 5.5 = 7.7 = 1 both groups non-cyclic so isomorphic <u>as only two groups of order 4</u> ALT Table is: <table><tr><td></td><td>1</td><td>3</td><td>5</td><td>7</td></tr><tr><td>1</td><td>1</td><td>3</td><td>5</td><td>7</td></tr><tr><td>3</td><td>3</td><td>1</td><td>7</td><td>5</td></tr><tr><td>5</td><td>5</td><td>7</td><td>1</td><td>3</td></tr><tr><td>7</td><td>7</td><td>5</td><td>3</td><td>1</td></tr></table> Isomorphism: $1 \leftrightarrow 1, (3,5,7) \leftrightarrow$ any permutation of (5,7,11) or states that structure is same		1	3	5	7	1	1	3	5	7	3	3	1	7	5	5	5	7	1	3	7	7	5	3	1	M1 M1 A1 [3] M1 M1 A1	Can be seen in table Or give order of each element (condone omission of e) Or all elements in each group are self-inverse or all have corresponding orders (shown) Can use So isomorphic as both are V or K_4 or Klein (four-)group or the four-group	
	1	3	5	7																											
1	1	3	5	7																											
3	3	1	7	5																											
5	5	7	1	3																											
7	7	5	3	1																											



Exam Papers Practice Copyright © 2024 Exam Papers Practice	order 4. Alternatively, they could state that both groups were copies of the Klein group. An alternate answer which avoided this pitfall
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Question			Answer/Indicative content	Marks	Part marks and guidance																							
52		a	<table border="1"><tr><td>7</td><td>2</td><td>1</td><td>1</td><td>1</td><td>1</td><td>4</td></tr><tr><td>4</td><td>2</td><td>1</td><td>7</td><td>6</td><td>5</td><td>3</td></tr><tr><td>+</td><td>-</td><td>-</td><td>+</td><td>-</td><td>+</td><td>+</td></tr></table> H_0: population median difference = 0 H_1: population median difference $\neq 0$ $P = 4+7+5+3=19$ $Q = 1+2+6 = 9$ $T = 8$ $T_{crit} = 3; 8 > 3$ Do not reject H_0. Insufficient evidence of a difference in test scores	7	2	1	1	1	1	4	4	2	1	7	6	5	3	+	-	-	+	-	+	+	M1(A01.1) B1(A02.5) A1(A03.3) A1(A03.4) B1(A01.1) A1FT(A02.2b) [6]	Calculate differences, rank them and attach signs Hypotheses correctly stated P or Q correct Both P and Q seen, T correct Comparison with correct T_{crit} Correct conclusion, in context, acknowledge uncertainty FT their T but not their T_{crit}	Follow through with correct signs and ranks from incorrect differences SC3: Two-sample, max 3 / 6	
7	2	1	1	1	1	4																						
4	2	1	7	6	5	3																						
+	-	-	+	-	+	+																						
		b	Uses magnitude of differences	B1(A03.5b) [1]																								
			Total	7																								

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Question			Answer/Indicative content	Marks	Part marks and guidance		
53		a	(i) If $p \equiv 0$ or $3 \pmod{6}$ then it is a multiple of 3 If $p \equiv 0, 2$ or $4 \pmod{6}$ then it is even Hence $p \equiv \pm 1 \pmod{6}$	B1(A02.1) E1(A01.1) [2]	The case $p \equiv 0$ needs only be dealt with once	May be in the form $p = 6k + r$ ($r = 0$ to 5)	
		a	(ii) $p^2 - 1 = (6k \pm 1)^2 - 1 = 36k^2 \pm 12k$ $= 12k(3k \pm 1)$ If k is even then $24 \mid 12k$ If k is odd then $3k \pm 1$ is even and $24 \mid 12k(3k \pm 1)$	M1(A02.1) A1(A01.1) E1(A02.4) [3]	Must be in a suitable form to make following deductions All fully explained	OR M1A1 Calculate all squares mod 24 (M1 if no more than 2 independent errors) E1 Explain that those congruent to 1 are $\pm 1, \pm 5, \pm 7, \pm 11$ i.e. all those $\equiv \pm 1 \pmod{6}$, of which the primes are a subset	
		b	$2^{p-1} = 2^{(p-1)(p+1)}$ $= (2^{p-1})^{p+1}$ $\equiv (1)^{p+1} \pmod{p}$ $= 1$	M1(A03.1a) M1(A03.1a) E1(A02.1) A1(A03.2a) [4]	By Fermat's little theorem, since $\text{hcf}(2, p) = 1$		



Question			Answer/Indicative content	Marks	Part marks and guidance		
		c	$p^{q-1} \equiv 1 \pmod{q}$ $q^{p-1} \equiv 0 \pmod{q}$ $\Rightarrow p^{q-1} + q^{p-1} \equiv 1 \pmod{q}$ Similarly, $p^{q-1} + q^{p-1} \equiv 1 \pmod{p}$ Together, $p^{q-1} + q^{p-1} \equiv 1 \pmod{pq}$	B1(A01.1) B1(A01.1) B1(A03.1a) B1(A03.1a) E1(A02.2a) [5]	By FLT (first result, either way round) Second result Since $\text{hcf}(p, q) = 1$	Must be justified	
			Total	14			



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Question			Answer/Indicative content	Marks	Part marks and guidance		
54		a	$(i) \begin{pmatrix} a & -b \\ b & a \end{pmatrix} \begin{pmatrix} c & -d \\ d & c \end{pmatrix} = \begin{pmatrix} ac-bd & -(ad+bc) \\ ad+bc & ac-bd \end{pmatrix}$ i.e. $p = ac - bd$ and $q = ad + bc$	M1(A01.1) A1(A01.1) [2]	May be clear from the product matrix		



Question			Answer/Indicative content	Marks	Part marks and guidance	
	a	(ii)	Suppose that both $p = 0$ and $q = 0$, so that $ac - bd = 0$ and $bc + ad = 0$	B1(A01.1)	Explicit assumption that both are zero	OR B1 Suppose that both $p = 0$ and $q = 0$ so that
			Then $acd - bd^2 = 0$ and $acd + bc^2 = 0$		Condition that a and b not both zero and c and d not both zero	$= 0$ so that $\det \begin{pmatrix} p & -q \\ q & p \end{pmatrix} = p^2 + q^2 = 0$
			Subtracting $\Rightarrow b(c^2 + d^2) = 0$	M1(A03.1a)	need not be explicitly stated	M1 Then $\det \begin{pmatrix} a & -b \\ b & a \end{pmatrix} \begin{pmatrix} c & -d \\ d & c \end{pmatrix} = 0$
			Since c and d are not both zero $c^2 + d^2 \neq 0 \Rightarrow b = 0$	A1(A02.2a)	e.g. treating as simultaneous equations	A1 so that $c^2 + d^2 = 0$ and / or $a^2 + b^2 = 0$
			Similarly, $ac - bd = 0$ and $bc + ad = 0 \Rightarrow a(c^2 + d^2) = 0$	A1(A02.2a)	(either a, b or c, d)	A1 since neither $\{c$ and $d\}$ or $\{a$ and $b\}$ can both be zero...
			Since c and d are not both zero $c^2 + d^2 \neq 0 \Rightarrow a = 0$	E1(A02.4)	One case correctly concluded	E1 This contradicts the condition that a, b and c, d are not both zero, so the assumption that p and q are both zero is false.
			This contradicts the condition that a, b are not both zero, so the assumption that p and q are both zero is false	[5]	Second case correctly concluded	
					Contradiction clearly explained	

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Question			Answer/Indicative content	Marks	Part marks and guidance		
		b	Closure follows from part (a) (Associativity is given) Identity is when $a = 1, b = 0$ or noting $\begin{pmatrix} 1 & (-)0 \\ 0 & 1 \end{pmatrix} \in X$ Inverse of $\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$ is $\frac{1}{a^2 + b^2} \begin{pmatrix} a & -(-b) \\ -b & a \end{pmatrix} \in X$ since $a^2 + b^2 \neq 0$ (as a, b not both zero)	B1(A01.1) B1(A01.1) B1(A02.1) E1(A02.3) [4]	To earn this, they must show appropriate form for A^{-1} and note why it exists		
		c	The matrix with $a = \cos\left(\frac{2}{17}\pi\right)$, $b = \sin\left(\frac{2}{17}\pi\right)$ generates ... the subgroup of rotations about O (through multiples of $\frac{2}{17}\pi$)	B1(A03.1a) B1(A02.4) [2]	Suitable values of a and b Description of the subgroup		
			Total	13			



Question			Answer/Indicative content	Marks	Part marks and guidance	
55		a	(i) $N_{t+1} - N_t = \frac{1}{750} N_t - \frac{1}{750} N_t^2 \Rightarrow \frac{1}{750} N_t (150 - N_t)$ Let M be a steady state value, then $\frac{1}{750} M (150 - M) = 0$ $M = 0$ or 150	B1(A03.4) M1(A02.1) A1(A01.1) [3]		
		a	(ii) $N_{t+1} - N_t = \frac{1}{750} N_t (150 - N_t)$ $N_t \in (0, 150) \Rightarrow \frac{1}{750} N_t \in (0, \frac{1}{5})$ Therefore, since $150 - N_t > 0$ We have that $N_{t+1} - N_t < 150 - N_t$	M1(A01.1) M1(A02.1) E1(A02.2a) [3]	soi	
		a	(iii) N_t increases to approach 150 in the long term, without oscillation	B1(A03.4) B1(A03.4) [2]	For "150" For "without oscillation" oe	
		b	(i) For Model I $N_4 = 15.82...$ For Model II $N_4 = 13$	B1(A01.1) B1(A01.1) [2]	BC Must show a non-integer answer BC	Using INT or by rounding after each iteration



Question			Answer/Indicative content	Marks	Part marks and guidance	
		b	(ii)E.g. The number of breeding pairs must be an integer, so Model II is the more appropriate model because it gives only integer values.	E1(A03.5a) [1]	For considering the appropriateness of the models given that the context requires N_t to be an integer.	
			Total	11		



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Question			Answer/Indicative content	Marks	Part marks and guidance		
56		a	(i) $f_x = 4x + 3y$ and $f_y = -2y + 3x + 17$ Both are zero when $4x + 3y = 0$ and $-2y + 3x + 17 = 0$ $x = -3, y = 4$ $z = 34$	B1(A01.1) B1(A01.1) M1(A02.1) A1(A01.1) A1(A01.1) [5]	The connection must be made explicitly		
		a	(ii) $f_{xx} = 4, f_{yy} = -2, f_{xy} = f_{yx} = 3$ $ H = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix} = \begin{vmatrix} 4 & 3 \\ 3 & -2 \end{vmatrix} = -17$ $ H < 0 \Rightarrow$ saddle-point	M1(A02.1) A1(A01.1) E1(A02.2a) [3]	Evaluating all second partial derivatives and attempt to use the Hessian FT from their above values Correct conclusion from a negative determinant value		
		b	$z = f(a, b) + (x - a)f_x(a, b) + (y - b)f_y(a, b)$ i.e. $z = 38 + 10(x - 1) + 16(y - 2)$ or $10x + 16y - z = 4$	M1(A01.1) A1(A01.1) [2]	Quoting correct form, including attempt to substitute in values Any correct simplified form		
Total				10			



Question			Answer/Indicative content	Marks	Part marks and guidance	
57		a	DR $I_n = \int_0^\pi \sin \theta \cdot \sin^{n-1} \theta \, d\theta$ $= \left[-\cos \theta \cdot \sin^{n-1} \theta \right]_0^\pi -$ $\dots \int_0^\pi -\cos \theta \cdot (n-1) \sin^{n-2} \theta \cdot \cos \theta \, d\theta$ $= 0 + (n-1) \int_0^\pi (1 - \sin^2 \theta) \sin^{n-2} \theta \, d\theta$ $= (n-1) \{I_{n-2} - I_n\}$ $\Rightarrow I_n = \frac{n-1}{n} I_{n-2}$	M1(A02.1) A1(A01.1) M1(A02.1) A1(A01.1) E1(A01.1) [5]	Correct splitting and attempt at $\int n$ by parts Use of trigonometric identity in order to obtain substitution for I AG	
		b	(i) DR $I_1 = \int_0^\pi \sin \theta \, d\theta$ $= 2$	M1(A03.1a) A1(A01.1) [2]		
		b	(ii) DR Using $c^2 = 1 - s^2$, $I = I$ $I_5 - I_7 = \left(1 - \frac{6}{7}\right) I_5 = \frac{1}{7} \times \frac{4}{5} I_3$ $= \frac{1}{7} \times \frac{4}{5} \times \frac{2}{3} I_1 = \frac{16}{105}$	M1(A03.1a) E1(A02.1) [2]	Repeated use of RF soi Correct answer with sufficient detail to show the reduction process	
			Total	9		



Question			Answer/Indicative content	Marks	Part marks and guidance		
58		a	Characteristic Equation is $\lambda^2 - 4\lambda + 4 = 0$ $\Rightarrow \lambda = 2$ (twice) and General Solution is $u_n = (A + Bn) \times 2^n$ $u_0 = 1$ and $u_1 = 1 \Rightarrow 1 = A$ and $1 = 2(A + B)$ $\Rightarrow A = 1, B = -\frac{1}{2}$ and $u_n = (1 - \frac{1}{2}n) \times 2^n$	M1(A01.1a) A1(A01.1) M1(A01.1) A1 FT(A01.1) [4]	Substitution of first two known terms FT correctly solved sim. eqns.		
		b	So $u_n = (2 - n) \times 2^{n-1}$ and both parts of the product are integers so u_n is an integer	M1(A03.1a) E1(A02.4) [2]		Do not penalise lack of checking for $u_0 = 1$ since this is given	
			Total	6			



Question			Answer/Indicative content	Marks	Part marks and guidance	
59			$\frac{\partial z}{\partial x} = \sin y - y \cos x$ $\frac{\partial z}{\partial y} = x \cos y + \cos x$ $\frac{\partial^2 z}{\partial x^2} = -y \cos x$ $\frac{\partial^2 z}{\partial y^2} = -x \sin y$ $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} + z$ $= -y \cos x - x \sin y + (x \sin y + y \cos x) = 0$	M1(A02.1) A1(A01.1) A1 FT(A01.1) A1 FT(A01.1) E1(A02.1) [5]	Attempt at 1 st and 2 nd partial derivatives of z with respect to x and y; at least one correct FT for each correct second partial derivative AG Shown clearly	
			Total	5		
60			$V = \frac{1}{6} \mathbf{a} \cdot \mathbf{b} \times \mathbf{c}$ $= \frac{1}{6} \times \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} -12 \\ 21 \\ -18 \end{pmatrix} = 3.5$	M1(A01.1a) A1(A01.1) A1 FT(A01.1) [3]	Attempt to use the triple scalar product A correct vector product BC BC FT their vector product if seen	
			Total	3		



Question			Answer/Indicative content	Marks	Part marks and guidance		
61			$\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = \left(2t - \frac{2}{t}\right)^2 + 16$ $A = 2\pi \int_1^4 4t \sqrt{\left(2t - \frac{2}{t}\right)^2 + 16} dt$ $= 384\pi \text{ so } k = 384$	M1(A01.1a) A1(A01.1) M1(A01.2) A1(A01.1) [4]	Correct unsimplified Use of Surface Area formula BC	May be seen in their expression for A. Must be seen	
			Total	4			



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Question			Answer/Indicative content	Marks	Part marks and guidance	
62		i	6 27	B1	Obtain correct values	
		i	129	B1	Obtain 3 rd correct value <u>Examiner's Comments</u> Most candidates found the first 3 terms correctly and deduced a correct value for (ii) from their values.	
		ii	3	B1ft	State a correct value	
		iii	$5^{n+1} + 2^n$	B1	Correct expression for u_{n+1} seen	Any letter, usually k or n seen
		iii		M1	Attempt to factorise $u_{n+1} + u_n$	Must deal with powers of 5 and 2
		iii		A1	Obtain correct simplified answer	e. g. $6 \times 5^n + 3 \times 2^n - 1$
		iii		A1	Clear explanation why u_{n+1} is divisible by 3	Not $u_{n+1} + u_n$ divisible by 3
		iii		B1	Clear statement of induction process <u>Examiner's Comments</u> Some candidates tried to prove directly that u_{n+1} has a divisor of 3, with limited success. Those who showed that $u_{n+1} + u_n$ has a divisor of 3, then failed to explain clearly how this led to u_{n+1} having a divisor of 3.	Provided other 4 marks earned
			Total	8		



Question			Answer/Indicative content	Marks	Part marks and guidance	
63		i	$\frac{2}{3}, \frac{2}{5}, \frac{2}{7}$	B1	B1 × 3, Obtain 3 correct values	
		i		B1		
		i		B1	Justify given answer Examiner's Comments Most found the first two answers correctly, but too frequently no working was shown to justify the value for u_4 .	
		ii	$\frac{2}{2n-1}$	M1	Fraction, in terms of n , with correct numerator or denominator	
		ii		A1	Obtain correct answer a.e.f. Examiner's Comments A correct expression, in terms of n , was usually seen, but a minority gave an expression in terms of u_n .	
		iii	$\frac{2}{2(n+1)-1}$	B1ft	Verify result true when $n = 1$, for their (ii), or $n = 2, 3$ or 4	
		iii		M1	Expression for u_{n+1} using recurrence relation in terms of n using their (ii)	
		iii		A1	Correct unsimplified answer	
		iii		A1	Correct answer in correct form	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii		B1	Specific statement of induction conclusion, previous 4 marks must be earned, $n = 1$ must be verified Examiner's Comments Sufficient working to demonstrate that the result was true for $n = 1$ was generally seen, as was the working to derive the correct expression for u_{k+1} from the recurrence relation and their part (ii). However, candidates should be made aware that the work required in an induction question is mainly of the "given answer" type, so sufficient justification is essential. Many candidates did not give a satisfactory conclusion to the induction process.	
			Total	10		



Question			Answer/Indicative content	Marks	Part marks and guidance	
64		i	$\forall n, 1n = n1 = n$ so 1 is identity	M1	Identify identity	can be implicit for M1
		i	But not all integers have an inverse, e.g. $2n \neq 1$ for any n	A1	Complete argument (example or general)	Examiner's Comments Clear arguments were key to gaining the marks in this part. The best answers first explicitly identified the identity as 1, and then either used a single counter-example to show that the set did not contain all inverses, or generalised by considering the inverse of a, for $a \neq 1, -1$. Those who just said that $\frac{1}{a}$ is not an integer did not gain the second mark, unless they demonstrated a fuller understanding by at least excluding $a = 1$. One misconception, which often escaped penalty due to sufficient correct reasoning, was the belief by some candidates that a lack of inverses within the set also demonstrated set closure.
	Copyright © 2024 Exam Papers Practice	ii	$\{-1, 1\}$ Demonstrates closure, references associativity references identity	B1*	...without contradiction	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$(-1)^{-1} = -1$ (and $1^{-1} = 1$) so inverses	* B2	B1 for any two of these Dep on 1 st B1	Examiner's Comments Many candidates answered this very well, correctly identifying $\{1, -1\}$ and itemising the four requirements for a group, closure usually being shown by a table, and the inverse of -1 being stated as -1. Historically, candidates sitting this paper have not always appeared aware of standard set notation, but this year it was rare to see sets described without the use of the correct style of parentheses. There was, however, still some confusion between associativity and commutativity. The most common error was to state that the group was $\{1\}$
			Total	5		



Question			Answer/Indicative content	Marks	Part marks and guidance	
65		i	$ba = a^n \Rightarrow b = a^{n-1}$	M1		Examiner's Comments Strong answers either used proof by contradiction, or considered each value of n and justified their rejection. These answers were marked by conciseness and a statement/s concerning the distinctness of the set elements defined by the question. Some otherwise good answers lost the final mark by omitting to conclude their argument with "So $ba \neq a^n$". Many candidates missed the point and talked only about what happens when you multiply a by itself repeatedly.
		i	But these are distinct elements so $ba \neq a^n$	A1		
		ii	$ba = a^2b$ $\Rightarrow a^2ba = a^4b$ $\Rightarrow a^2ba = b$	M1	$or\ b = a^2ba^3, a = ba^2b, a^2 = bab\ or\ b = ba^2$	
		ii	$\Rightarrow a^2ba^4 = ba^3$ $\Rightarrow a^2b = ba^3$ $\Rightarrow ba = ba^3$ $\Rightarrow e = a^2$	M1	validly reach any equality which gives 2 distinct elements of the group as equal	
		ii	Which is false, hence $ba \neq a^2b$	A1	Complete argument	
		ii	If $ba = ab$ then (all element pairs would have to be commutative and so) G would be abelian.	M1	Do not award for G non-abelian $\Rightarrow ba \neq ab$	
		ii	If $ba = b$ then $a = e$ so $ba \neq b$.	M1		



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	So, by elimination of other possibilities, $ba = a^3b$	A1	Dependent on all previous marks	Examiner's Comments This question proved beyond the capabilities of most candidates, and although all the different marks available were accessed by some candidates, it was rare to see them all combined to form a complete proof. Good answers to the proof that $ba \neq a^2b$ came from candidates who persevered to find a contradiction. Most candidates tried to prove that ba was not equal to ab by element operation, or presumed that the non-commutative nature of the group meant that no two elements in the set commuted with each other, rather than using the fact that if ab were equal to ba then all element pairs would consequently have to be commutative. Many candidates sensibly attempted to deduce that $ba = a^3b$ by eliminating other options, but most failed to notice that they had to rule out b as a possible candidate for ba.
		iii	$ba^2 = baa = a^3ba$	M1	Use previous result	



Question			Answer/Indicative content	Marks	Part marks and guidance																										
		iii	$= a^3a^3b = a^2b$	A1	Complete argument	Examiner's Comments Good answers to this question started with one side of the identity to be proved and used a clear chain of steps linked by equals signs to the other side of the identity. Many candidates demonstrated poor mathematical language by starting with b $a^2 = a^2b$, and continuing through steps to show that this implied that $a^2b = a^2b$, or even worse that $e = e$. Some candidates unsuccessfully attempted to show this identity without using the fact that $ba = a^3b$.																									
		iv	<table><tr><td></td><td>e</td><td>a</td><td>a^2</td><td>a^3</td></tr><tr><td>e</td><td>e</td><td>a</td><td>a^2</td><td>a^3</td></tr><tr><td>a</td><td>a</td><td>a^2</td><td>a^3</td><td>e</td></tr><tr><td>a^2</td><td>a^2</td><td>a^3</td><td>e</td><td>a</td></tr><tr><td>a^3</td><td>a^3</td><td>e</td><td>a</td><td>a^2</td></tr></table>		e	a	a^2	a^3	e	e	a	a^2	a^3	a	a	a^2	a^3	e	a^2	a^2	a^3	e	a	a^3	a^3	e	a	a^2	B1	All correct	
	e	a	a^2	a^3																											
e	e	a	a^2	a^3																											
a	a	a^2	a^3	e																											
a^2	a^2	a^3	e	a																											
a^3	a^3	e	a	a^2																											
		iv	<table><tr><td></td><td>e</td><td>a^2</td><td>b</td><td>a^2b</td></tr><tr><td>e</td><td>e</td><td>a^2</td><td>b</td><td>a^2b</td></tr><tr><td>a^2</td><td>a^2</td><td>e</td><td>a^2b</td><td>b</td></tr><tr><td>b</td><td>b</td><td>a^2b</td><td>e</td><td>a^2</td></tr><tr><td>a^2b</td><td>a^2b</td><td>b</td><td>a^2</td><td>e</td></tr></table>		e	a^2	b	a^2b	e	e	a^2	b	a^2b	a^2	a^2	e	a^2b	b	b	b	a^2b	e	a^2	a^2b	a^2b	b	a^2	e	B1	Correct elements	
	e	a^2	b	a^2b																											
e	e	a^2	b	a^2b																											
a^2	a^2	e	a^2b	b																											
b	b	a^2b	e	a^2																											
a^2b	a^2b	b	a^2	e																											
		iv		M1	At least 12 out of 16 entries correct																										
		iv		A1	All correct																										
		iv	<table><tr><td></td><td>e</td><td>a^2</td><td>ab</td><td>a^3b</td></tr><tr><td>e</td><td>e</td><td>a^2</td><td>ab</td><td>a^3b</td></tr><tr><td>a^2</td><td>a^2</td><td>e</td><td>a^3b</td><td>ab</td></tr><tr><td>ab</td><td>ab</td><td>a^3b</td><td>e</td><td>a^2</td></tr><tr><td>a^3b</td><td>a^3b</td><td>ab</td><td>a^2</td><td>e</td></tr></table>		e	a^2	ab	a^3b	e	e	a^2	ab	a^3b	a^2	a^2	e	a^3b	ab	ab	ab	a^3b	e	a^2	a^3b	a^3b	ab	a^2	e	B1	Correct elements	
	e	a^2	ab	a^3b																											
e	e	a^2	ab	a^3b																											
a^2	a^2	e	a^3b	ab																											
ab	ab	a^3b	e	a^2																											
a^3b	a^3b	ab	a^2	e																											



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv		M1	At least 12 out of 16 entries correct	Examiner's Comments There was evidence that most candidates had some familiarity with the structure of groups of order 4 in that most were able to gain some marks here. It is clear that some candidates have been well drilled in the cyclic and Klein patterns and were able to gain most marks here even when they had been unsuccessful with earlier parts of the question. The cyclic table was most often found, although even this occasionally had incorrect elements within it. Many also found at least one of the two Klein groups $\{e, a^2, b, a^2 b\}$ and $\{e, a^2, ab, a^3 b\}$ although it was not uncommon for the leading diagonal to mistakenly contain elements other than the identity element.
		iv		A1	All correct	
			Total	17		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
66		i	0	B1	accept $0 + 0x + 0x^2$ <u>Examiner's Comments</u> This year there were fewer candidates who appeared completely unfamiliar with groups, although this is still the topic that causes the greatest difficulties. Almost all candidates were successful in this part.	
		ii	$2 + 3x + 4x^2$	M1	for 2 correct terms	
		ii		A1	<u>Examiner's Comments</u> This year there were fewer candidates who appeared completely unfamiliar with groups, although this is still the topic that causes the greatest difficulties. Almost all candidates were successful in this part but those that failed to score marks usually assumed, erroneously, that negative values exist within the group, giving an answer of $-3 - 2x - x^2$	
		iii	125	B1	or 5^3 <u>Examiner's Comments</u> This part was often answered correctly but the following incorrect values were regularly seen: 5, 25 and 60 (from 5P3) and 120 (from 5!)	
		iv	more than five elements are shown to be generated so $ H > 5$		e.g. elements generated by $1 + x$ are $\{1 + x, 2 + 2x, 3 + 3x, 4 + 4x, 0\}$	
		iv		B1	which does not include $2 + x$	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	$ H $ is a factor of 125	B1	or order subgroups 1, 5, 25 or 125	Insufficient to just reference Lagrange alone
		iv	proper so $ H < 125$	B1	or order is (1), 5, 25	
		iv	$ H = 25$	B1	<u>Examiner's Comments</u> This part was a good differentiator between candidates. Good answers were those that carefully justified each step required to reach the conclusion that the order was 25. Some candidates were not sufficiently clear in showing that they knew what distinguished proper subgroups from trivial ones. A common alternative approach was to claim, with varying degrees of corroboration, that H consisted of all elements of the form $a+bx$. Unfortunately, most candidates taking this approach were unable to adequately justify it; often they assumed, without substantiating it, that the two given elements formed a generating set. Some candidates demonstrated poor conceptual understanding of "order" claiming that the order of the group was 5 because both $1+x$ and $2+x$ were of order 5.	penalise use of H instead of $ H $
			Total	8		



Question			Answer/Indicative content	Marks	Part marks and guidance	
67		i	$eg = ge$ so $e \in H$	B1	Showing identity in H	
		i	$hg = gh$			
		i	$\Rightarrow g = h^{-1}gh$	M1		
		i	$\Rightarrow gh^{-1} = h^{-1}g$	M1		
		i	$\Rightarrow h^{-1} \in H$	A1		
		i	$h_1h_2g = h_1gh_2$	M1		
		i	$= gh_1h_2$	M1		
		i	so $h_1h_2 \in H$ so H closed	A1		
		i	so H is a subgroup of G	A1	For completing argument without considering other properties of H	
					<u>Examiner's Comments</u> The best answers to this question were marked by a clear understanding of what needed to be shown, not just in terms of the basic group axioms, but particularly in terms of the fact that these axioms had to be shown to be true for elements that obeyed the condition for membership of the subgroup. The very best candidates were able to demonstrate closure of subset, existence of both inverses and identity within subset. They also knew that it was trivial that subsets of a group are associative. They showed this with clear arguments that didn't confuse the general with the particular. As in previous years there was clear evidence that candidates had studied the topic and knew the four group axioms; however,	



Question			Answer/Indicative content	Marks	Part marks and guidance	
					they frequently had no concept of how to apply these in context. Many candidates assumed closure within the group or the existence of an inverse within the group automatically led to the same in the subgroup. However, they then went to lengths to try to show associativity within the subgroup when this is genuinely inherited. Many were confused by the term 'multiplicative group' thinking that this meant that they were dealing with the algebra of real numbers. This led them to erroneously equate the identity to 1 and the inverse $\frac{1}{h}$ to $\frac{1}{h}$	
		ii	Correctly evaluates first $g_1 g_2$	B1*		where g_1, g_2 distinct and $\neq e$
		ii	$g_1 g_2 \neq g_2 g_1$ for one correct pair	M1		
		ii	$g_1 g_2 \neq g_2 g_1$ for sufficient pairs to cover all 5 elements and conclude that they are not in H	A1		



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	so $H = \{e\}$	A1dep*	Complete argument <u>Examiner's Comments</u> Once again, it was clarity of mathematical argument that marked out the very best answers here. Good answers systematically eliminated elements (or potential subgroups) by counterexample. Some lost a mark because they incorrectly evaluated g_1g_2 as g_2g_1 but otherwise knew how to tackle the problem. Others only considered some of the elements or talked in vague terms about the non-symmetrical nature of the combination table.	
			Total	12		



Question			Answer/Indicative content	Marks	Part marks and guidance																			
68		i	<table><tr><td>element</td><td>(1)</td><td>3</td><td>7</td><td>9</td><td>11</td><td>13</td><td>17</td><td>19</td></tr><tr><td>inverse</td><td>(1)</td><td>7</td><td>3</td><td>9</td><td>11</td><td>17</td><td>13</td><td>19</td></tr></table>	element	(1)	3	7	9	11	13	17	19	inverse	(1)	7	3	9	11	17	13	19	B1	2 or more	Ignore 1
	element	(1)	3	7	9	11	13	17	19															
	inverse	(1)	7	3	9	11	17	13	19															
		i		B1	4 or more																			
	i		B1	all 7 correct																				
				Examiner's Comments It was good to see better use of braces { } to define sets but few candidates knew formal mathematical notation for equivalence when stating an isomorphism. Centres may wish to draw candidates' attention to the list of Mathematical notation given in appendix B of the specification. (Once again poor notation was not, this year, penalised here.) A few candidates had very poor basic understanding this topic; they were either unfamiliar with modulo multiplication, or unable to identify the order of an element or of a group. Most, however, were able to access the question well. This was standard work for most candidates.																				
	© 2024, Exam Papers Practice	ii	(1 has order 1) 9,11,19 have order 2	M1	Correctly identifies order of all elements	Allow one error																		
		ii	$3^2 = 9 \Rightarrow 3^4 = 1$ so order 4 similarly 7,13,17 order 4	B1	justifies order for at least 1 element of order 4	must show workings towards a^4 for demonstration that these elements are order 4																		



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	no element of order 8 so not cyclic	A1	www Examiner's Comments It was good to see better use of braces { } to define sets but few candidates knew formal mathematical notation for equivalence when stating an isomorphism. Centres may wish to draw candidates' attention to the list of Mathematical notation given in appendix B of the specification. (Once again poor notation was not, this year, penalised here.) A few candidates had very poor basic understanding this topic; they were either unfamiliar with modulo multiplication, or unable to identify the order of an element or of a group. Most, however, were able to access the question well. In general, candidates understood how to show that G was not cyclic by considering the order of each element and of the group, but some failed to give sufficient detail to their demonstration, with many not showing clearly why elements such as 3 had order 4. Others neglected to say what they had shown in conclusion.	condone "no generator" in place of "no element of order 8"
		iii		M1	For two sets which both contain "1" and all (4) elements' inverses	
		iii		B1	One subgroup of order 4	
		iii	{1,13, 9, 17} and {1, 3, 9, 7}	A1		



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii		M1	for correspondence of “their” elements of same order	
		iii	$1 \leftrightarrow 1, 9 \leftrightarrow 9, 3 \leftrightarrow 13, 7 \leftrightarrow 17$	A1	or $3 \leftrightarrow 17, 7 \leftrightarrow 13$	
					Examiner's Comments	
					It was good to see better use of braces { } to define sets but few candidates knew formal mathematical notation for equivalence when stating an isomorphism. Centres may wish to draw candidates' attention to the list of Mathematical notation given in appendix B of the specification. (Once again poor notation was not, this year, penalised here.) A few candidates had very poor basic understanding this topic; they were either unfamiliar with modulo multiplication, or unable to identify the order of an element or of a group. Most, however, were able to access the question well.	
					{1, 3, 7, 9} was generally found to be a subgroup of order 4 by most candidates but the second set of elements was far more varied and did not always constitute a group either due to the absence of inverse elements or due to lack of closure. Many failed to comprehend the request to “state an isomorphism between (the subgroups)”, but those that did usually understood the need for equivalent elements to be of the same order. Whilst it was acceptable for	



Question			Answer/Indicative content	Marks	Part marks and guidance	
					candidates to give two alternative complete mappings, this needed to be precisely explained and those who gave “3 □ 13 or 17, and 7 □ 13 or 17” were not awarded the final accuracy mark.	
			Total	11		



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Question			Answer/Indicative content	Marks	Part marks and guidance	
69		i	$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ is identity	B1		
		i	$\begin{pmatrix} a & -b \\ b & a \end{pmatrix}^{-1} = \frac{1}{a^2 + b^2} \begin{pmatrix} a & b \\ -b & a \end{pmatrix} \in G$	M1	for M1, must at least get matrix in form $\begin{pmatrix} x & -y \\ y & x \end{pmatrix}$, or state existence of inverse due to non-singularity	
		i		A1		
		i	$\begin{pmatrix} a & -b \\ b & a \end{pmatrix} \begin{pmatrix} c & -d \\ d & c \end{pmatrix} = \begin{pmatrix} ac - bd & -bc - ad \\ bc + ad & ac - bd \end{pmatrix}$	M1		
		i	and $(ac - bd)^2 + (bc + ad)^2 = a^2 c^2 + b^2 d^2 + b^2 c^2 + a^2 d^2$	M1		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
		i	$= (a^2 + b^2)(c^2 + d^2) \neq 0$	A1	Must not attempt to prove commutativity in (i) Examiner's Comments As in question 4, a few candidates failed to access this question due to their lack of knowledge of matrix algebra, or of basic set vocabulary. The majority of candidates knew the basic axioms for a group, although there did seem some lack of awareness that elements not only have to have an inverse, but that the said inverse must be contained within the group. So the inverse was often found or stated but rarely was it explicitly said to be within the group. Closure was considered in most cases, but with extremely few considering the required condition on the determinant. Some candidates consider $a \circ a$ rather than $a \circ b$ when exploring closure.	
	Copyright © 2024 Exam Papers Practice	ii	$\begin{pmatrix} c & -d \\ d & c \end{pmatrix} \begin{pmatrix} a & -b \\ b & a \end{pmatrix} = \begin{pmatrix} ac-bd & -bc-ad \\ bc+ad & ac-bd \end{pmatrix}$	M1		must also consider matrices reversed, but may be seen in (i)
		ii	$= \begin{pmatrix} a & -b \\ b & a \end{pmatrix} \begin{pmatrix} c & -d \\ d & c \end{pmatrix}$ so commutative	A1		



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii			Examiner's Comments As in question 4, a few candidates failed to access this question due to their lack of knowledge of matrix algebra, or of basic set vocabulary. This question and part (iii) were generally well answered. The most common errors made were: not using general matrices, using matrices not in the form of elements of the group or attempting to show associativity in place of commutativity.	
		iii	$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}^2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$	M1	g^2 must be correct	
		iii	$\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$	M1	allow 1 error in getting g^4	
		iii	order 4	A1		
		iii			Examiner's Comments As in question 4, a few candidates failed to access this question due to their lack of knowledge of matrix algebra, or of basic set vocabulary. Although most candidates tackling this question did all the calculations correctly, some then claimed that the order of this element was 3 or 5.	
			Total	11		



Question			Answer/Indicative content	Marks	Part marks and guidance																										
70		i	<table><tr><td></td><td>1</td><td>3</td><td>5</td><td>7</td></tr><tr><td>1</td><td>1</td><td>3</td><td>5</td><td>7</td></tr><tr><td>3</td><td>3</td><td>1</td><td>7</td><td>5</td></tr><tr><td>5</td><td>5</td><td>7</td><td>1</td><td>3</td></tr><tr><td>7</td><td>7</td><td>5</td><td>3</td><td>1</td></tr></table>		1	3	5	7	1	1	3	5	7	3	3	1	7	5	5	5	7	1	3	7	7	5	3	1	B2	-1 each error	
		1	3	5	7																										
	1	1	3	5	7																										
3	3	1	7	5																											
5	5	7	1	3																											
7	7	5	3	1																											
	i	From table clearly closed	B1			Must be clear they are referring to tabulated results																									
	i	1 is identity	B1			Or “1 appears in every row”																									

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Question			Answer/Indicative content	Marks	Part marks and guidance	
		i	$3^{-1} \equiv 3, 5^{-1} = 5, 7^{-1} \equiv 7 \pmod{8}$	B1	Superfluous fact/s gets -1 Examiner's Comments The first three parts of this question were done well but part (iv) less well. However here, as elsewhere, candidates lost marks though not giving sufficiently thorough justifications. This is especially important in group questions since candidates have to persuade the reader that they are actually considering the set given in the question. Virtually all candidates correctly formed the group table, and most knew the group axioms. A few evidently thought that any group has to be commutative. However the main cause of dropped marks was an inability to be convincing in demonstration. This was especially true when candidates addressed closure. A few candidates decided that closure required each row to contain every element of the set. More candidates simply did not convince the examiner that they were relating their conclusion to G, giving instead a general definition of closure or simply saying "it's closed".	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	1 has order 1 and 3, 5, 7 all have order 2	B1	Examiner's Comments A common error here was to state that the identity had the same order as the other elements.	
		iii	{1, 3}, {1, 5}, {1, 7} (and {1})	B1	All correct, no extras Examiner's Comments The main weakness observed in this part was poor set notation (which was not penalised this time). Candidates should be encouraged to use braces { } for sets.	Allow {1} included or omitted
		iv	in H $3^2 \equiv 9 \pmod{10}$ so 3 not order 2	M1	Shows and states that 3 or that 7 is not order 2 (or is order 4)	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	no element of order > 2 in G so not isomorphic	A1	Completely correct reasoning Or, if zero, then SC1 for merely stating comparable orders and then saying that “orders don’t correspond, so not isomorphic” Or table for H with saying “not all elements self inverse, so not isomorphic” Examiner’s Comments This was a question where the need for rigour appeared to be lost on many candidates. The arguments presented were often only partial. For instance candidates often compared the orders of elements, but did not fully justify claims that 3 (for example) has order 4. A few seemed to want to treat the element 3 as corresponding in G and H.	
			Total	9		



Question			Answer/Indicative content	Marks	Part marks and guidance	
71		i	$(ab)^6 = abab\dots ab = a^6b^6$ as commutative	M1	Must give reason	Some demonstration that they understand commutativity
		i	$= (a^2)^3 (b^3)^2 = e^3 e^2 = e$	A1	Using orders of a and b	
		i	So ab has order 1, 2, 3, or 6 ($b \neq a \Rightarrow ab \neq a^2 \Rightarrow ab \neq e$ so ab not order 1)			Condone absence of this line
		i	$(ab)^2 = a^2b^2 = eb^2 = b^2$ and b not order 2, so ab not order 2	M1	Consider other cases	Insufficient to merely have simplified all $(ab)^n$
		i	$(ab)^3 = a^3b^3 = aa^2e = aee = a \neq e$, so ab not order 3			
		i	(So must be order 6)	A1	AG Complete argument	
					Examiner's Comments This question proved challenging with only a few candidates demonstrating the ability to form cogent, logical arguments. Candidates need to positively show their reasoning rather than leaving the examiner to intuit it. This was generally poorly answered. There was seldom reference to commutativity – candidates cannot simply assume that $(ab)^6 = a^6b^6$. Amongst those who did show that $(ab)^6 = e$, there was seldom explicit consideration of why ab did not have order 2 or 3. Very few candidates gave a full answer to this.	
		ii	ac has order 18	B1		Or abc or generator



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	18 is LCM of 2 and 9, (so we can use a similar argument to part (i))	M1	or explicit consideration of other cases	
		ii	So as G has an element of order 18 it must be cyclic.	A1	AG Complete argument Examiner's Comments It was rare to see a full and concise reasoned solution using least common multiple. Quite a few candidates intuited that ac (or abc) has order 18, but not many justified this properly. Many that did either computed all powers up to the 18th, or computed the factor powers, often without referencing Lagrange's theorem.	
			Total	7		



Question			Answer/Indicative content	Marks	Part marks and guidance	
72		i	$I_n = \int_0^{\pi/4} \sec^n x \, dx$			
		i	$u = \sec^{n-2} x \, dv = \sec^2 x \, dx$	M1	Attempt at integration by parts with correct u and v'	
		i	$du = (n-2) \sec^{n-3} x \cdot \sec x \tan x \, dx \quad v = \tan x$			
		i	$\Rightarrow I_n = \left[\sec^{n-2} x \tan x \right]_0^{\pi/4}$	A1	" uv " term must be seen	
			$- \int_0^{\pi/4} (n-2) \sec^{n-3} x \cdot \sec x \tan x \cdot \tan x \, dx$			
		i	$= (\sqrt{2})^{n-2} - (n-2) \int_0^{\pi/4} \sec^{n-2} x \cdot \tan^2 x \, dx$	A1	Evaluating the first term	
		i	$= (\sqrt{2})^{n-2} - (n-2) \int_0^{\pi/4} \sec^{n-2} x (\sec^2 x - 1) \, dx$	M1	Dep on 1st M Splitting integral using	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		i	$= (\sqrt{2})^{n-2} - (n-2)(I_n - I_{n-2}) \Rightarrow$ $(n-1)I_n = (\sqrt{2})^{n-2} + (n-2)I_{n-2}$	A1	$\tan^2 x = \sec^2 x - 1$	Examiner's Comments Some candidates seemed to run out of time (or inspiration) but most made an attempt at this question. Most candidates realised that they needed to use integration by parts but some used $\sec^2 x = 1 + \tan^2 x$ before integrating, which made the problem much more difficult and with the wrong choice for “u” and “v” impossible. A small number of candidates split the parts correctly but were not able to differentiate $\sec^n - 2x$ correctly, often saying that $u' = (n-2)\sec^n - 3x \tan x$ instead of $(n-2)\sec^n - 3x \sec x \tan x$, leading to an expression for I_n that involved I_{n-1} and I_{n-3} instead of I_n and I_{n-2}.
		ii	$I_8 = \frac{1}{7}(2^3 + 6I_6) = \frac{1}{7}\left(2^3 + 6 \cdot \frac{1}{5}(2^2 + 4I_4)\right)$	B1	Correct statement of formula seen anywhere	
		ii	$= \frac{1}{7}\left(2^3 + 6 \cdot \frac{1}{5}\left(2^2 + 4 \cdot \frac{1}{3}(2 + 2I_2)\right)\right)$			
		ii	$= \frac{1}{7}\left(8 + \frac{6}{5}\left(4 + \frac{16}{3}\right)\right) = \frac{1}{7}\left(8 + \frac{6}{5} \cdot \frac{28}{3}\right)$	B1	Substitution of I_2 seen oe	
		ii	$= \frac{1}{7} \cdot \frac{96}{5} = \frac{96}{35}$	B1	Conclusion	
		ii	Alternative:		B1 for I_2 B1 For I_4 and I_6 B1	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$I_2 = 1 \quad I_4 = \frac{4}{3} \quad I_6 = \frac{28}{15} \Rightarrow I_8 = \frac{96}{35}$			Examiner's Comments Some candidates seemed to run out of time (or inspiration) but most made an attempt at this question. Whether the recurrence formula had been achieved or not, most candidates used the given result to connect I_8 to I_6, I_4 and I_2. The majority of the candidates were able to show that $I_2 = 1$ and hence deduce $I_4 = \frac{4}{3}$, $I_6 = \frac{28}{15}$ and $I_8 = \frac{96}{35}$
		iii	$I_2 = 1$ (which is rational). (Therefore I_{2n} is rational for $n = 1$).	B1	Allow $n = 2$	
		iii	Let I_{2k} be rational for some value of k Then $I_{2(k+1)} = \frac{1}{(2k+1)}(\sqrt{2}^{2k} + 2kI_{2k})$ $= \frac{1}{(2k+1)}(2^k + 2kI_{2k})$	M1	For statement plus reduction formula and begin to look at the $\sqrt{2}$ term	Assume true for $n = k$
		iii	Statement that this is rational and must include	A1		



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$\sqrt{2}^{2k} = 2^k$ is rational So if I_{2k} is rational then $I_{2(k+1)}$ is rational. But I_2 is rational so I_{2k} is rational for all $n = k$	A1	Reduction formula must be correct	Examiner's Comments Some candidates seemed to run out of time (or inspiration) but most made an attempt at this question. The initial case for the induction followed (since I_2, or I_4, is rational). The most successful candidates were those who set out their induction formally, stating that the assumption was that I_{2k} is rational and then using the recurrence formula to relate I_{2k+2} to I_{2k}, together while reaching the conclusion. Far too many candidates did not discuss the nature of the $\sqrt{2}^{2k}$ term. It was disappointing to see so many candidates write "assume true for $n = 2k$" but fail to state that they now needed to show that it was also true for $n = 2k + 2$ and wrote down the reduction formula from (i) for $n = 2k$ and this caused them confusion regarding what they needed to do.
			Total	12		



Question			Answer/Indicative content	Marks	Part marks and guidance	
73		i	$I_n = \int_0^1 x^n e^{-x} dx \quad u = x^n \quad dv = e^{-x} dx$ $du = nx^{n-1} dx \quad v = -e^{-x}$	M1	By parts	
		i	$I_n = \left[-e^{-x} x^n \right]_0^1 + n \int_0^1 x^{n-1} e^{-x} dx$ $= (-e^{-1} - 0) + nI_{n-1}$	A1	Both terms before limits are applied soi	
		i	$I_n = nI_{n-1} - e^{-1}$	A1	Or $k = \frac{-1}{e}$ Examiner's Comments This was a standard reduction formula, which was usually done well – 95% of candidates achieved full marks.	
		ii	$I_0 = \int_0^1 e^{-x} dx = \left[-e^{-x} \right]_0^1 = 1 - e^{-1}$	B1		Or finding I_1 . Or could be done the other way round.
		ii	$I_3 = 3I_2 - e^{-1}$ $= 3(2I_1 - e^{-1}) - e^{-1} = 6I_1 - 4e^{-1}$ $= 6(I_0 - e^{-1}) - 4e^{-1} = 6I_0 - 10e^{-1}$	M1	Complete method even if k is wrong.	
		ii	$I_3 = 6 - 16e^{-1}$	A1	SC3 by parts 2 or 3 times Examiner's Comments This was done less well but still a good source of marks for most candidates.	
	Copyright © 2024 Exam Papers Practice	iii	$I_{11} = 11I_{10} - e^{-1}$	M1	Complete method (Could be done the other way round.)	Alternative: Starting from I_4 and working up to I_{11} M1
		iii	$= 11(10I_9 - e^{-1}) - e^{-1} = 110I_9 - 12e^{-1}$	A1	For I_{11} in terms of I_{10} or I_9 in terms of I_8 soi	I_8 or I_{11} correct A1 $I_8 = 8! - \frac{109601}{e}, I_{11} = 11! - \frac{108505112}{e}$



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$=110(9I_8 - e^{-1}) - 12e^{-1} =$ $990I_8 - 122e^{-1}$ $990I_8 - I_{11} = 122e^{-1}$	A1	Examiner's Comments This was completed by only about 30% of candidates. Those that started with I_{11}, used the reduction formula three times found that $990 I_8$ was involved from which the answer came easily. A significant number thought that the demand to find I_3 in (ii) was a hint to keep going up to I_{11}. This involved a great deal of work with huge numbers and it was remarkable how many achieved the right result in the end. This method, of course, was a poor use of time given that there were only three marks available.	
			Total	9		



Question			Answer/Indicative content	Marks	Part marks and guidance	
74		i	$I_n = \int_0^1 x^n \cdot e^{2x} dx.$			
		i	Set $u = x^n$ $du = nx^{n-1} dx$	M1	Integration by parts	
		i	$dv = e^{2x} dx \quad v = \frac{1}{2} e^{2x}$	A1	Correct way round and correct diffn	
		i	$\Rightarrow I_n = \int_0^1 x^n e^{2x} dx = \left[\frac{1}{2} x^n e^{2x} \right]_0^1 - \frac{1}{2} n \int_0^1 x^{n-1} e^{2x} dx$ $I_n = \frac{1}{2} e^2 - \frac{1}{2} n I_{n-1}$	A1 A1	Indefinite form acceptable Using limits	
		i			<u>Examiner's Comments</u> This standard reduction formula was usually well done, though a few candidates jumped to the result rather too soon, missing essential algebraic steps.	
		ii	$I_0 = \int_0^1 e^{2x} dx = \frac{1}{2} [e^{2x}]_0^1 = \frac{1}{2} (e^2 - 1)$ $I_1 = \frac{1}{2} e^2 - \frac{1}{2} I_0 = \frac{1}{2} e^2 - \frac{1}{2} \left(\frac{1}{2} (e^2 - 1) \right) = \frac{1}{4} e^2 + \frac{1}{4}$ $I_2 = \frac{1}{2} e^2 - I_1 = \frac{1}{2} e^2 - \left(\frac{1}{4} e^2 + \frac{1}{4} \right) = \frac{1}{4} e^2 - \frac{1}{4}$ $I_3 = \frac{1}{2} e^2 - \frac{3}{2} I_2 = \frac{1}{2} e^2 - \frac{3}{2} \left(\frac{1}{4} e^2 - \frac{1}{4} \right) = \frac{1}{8} e^2 + \frac{3}{8}$	M1 A1 M1 A1	Attempt to find I_0 or I_1 . Using this to progress, dep	
		ii			<u>Examiner's Comments</u> This part was also done well. A few found I_1 initially rather than the easier I_0 . A very small number of candidates ignored part (i) completely and found I_3 by successive reductions.	
			Total	8		



Question			Answer/Indicative content	Marks	Part marks and guidance	
75		i	$I_n = \int_0^{\pi/2} \sin^n x \, dx = I_n = \int_0^{\pi/2} \sin^{n-1} x \sin x \, dx$	M1	Correct start for reduction	
		i	$\Rightarrow I_n = \left[\sin^{n-1} x (-\cos x) \right]_0^{\pi/2} + \int_0^{\pi/2} \cos x (n-1) \sin^{n-2} x \cos x \, dx$			
		i	$= 0 + (n-1) \int_0^{\pi/2} \sin^{n-2} x (1 - \sin^2 x) \, dx = (n-1)(I_{n-2} - I_n)$	M1	Deal with $\cos^2$ dep on 1st M	
		i	$\Rightarrow nI_n = I_{n-2} \Rightarrow I_n = \frac{n-1}{n} I_{n-2}$	A1	www Examiner's Comments Setting up the reduction formula in was quite a challenge for some candidates, either because they tried to set $\frac{dy}{dx} = 1$ instead of $\sin x$, or because they could not differentiate $u = \sin^{n-1} x$. The result was given, however, so this should not have prevented candidates from making progress with the other two parts of the question.	
		ii	$I_n = \frac{n-1}{n} I_{n-2} \Rightarrow I_{2n+1} = \frac{(2n+1)-1}{(2n+1)} I_{2n-1} = \frac{2n}{(2n+1)} I_{2n-1}$	M1	Allow using n instead of $2n + 1$ Examiner's Comments The algebra in was often not well done with candidates using expressions in n instead of $2n$ and not realising that the fraction $\frac{2n}{2n+1}$ had to be less than 1 for positive n .	Alt: M1 $y = \sin^n x < y = \sin^{n-2} x$ in range
		ii	and $\frac{2n}{(2n+1)} < 1$	A1		A1 means that the area underneath is less and therefore...



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	Alternative $I_n = \frac{n-1}{n} I_{n-2}, \frac{n-1}{n} < 1 \Rightarrow I_n < I_{n-2}$ for all $n \Rightarrow I_{2n+1} < I_{2n-1}$			This can be argued one step at a time instead of 2
		iii	$I_{11} = \frac{256}{693}, I_{10} = \frac{63}{512} \cdot \pi, I_9 = \frac{128}{315}$	B1	For I_1 soi	Allow for pa.
		iii		B1	For I_0 soi	
		iii		M1	Applying reduction formula for at least one of I_9 and I_{11}	
		iii	$\Rightarrow \frac{256}{693} < \frac{63}{512} \cdot \pi < \frac{128}{315}$	M1	Applying reduction formula for I_{10}	
		iii	$\Rightarrow \frac{131072}{43659} < \pi < \frac{65536}{19845}$	A1	Lhs fraction or decimal equivalent correct to 4dp	Both correct but both only to 3sf give A1 only
		iii	$\Rightarrow 3.0022 < \pi < 3.3024$	A1	Likewise Rhs	
					Examiner's Comments Candidates often gave up on having reached values for I_9 and I_{11} , at which point they had done most of the work but only earned 2 marks out of 6.	
			Total	11		



Question			Answer/Indicative content	Marks	Part marks and guidance	
76		i	$u = \cos^{n-1} x, dv = \cos x dx$	M1*	By parts the right way round	
		i	$du = -(n-1)\cos^{n-2}x \sin x, v = \sin x$	A1		
		i	$\Rightarrow I_n = \left[\cos^{n-1} x \sin x \right]_0^{\frac{\pi}{2}} + (n-1) \int_0^{\frac{\pi}{2}} \cos^{n-2} x \sin^2 x dx$	A1	Integral so far	
		i	$= 0 + (n+1)(I_{n-2} - I_n)$	*M1	Correct use of $\sin^2 x = 1 - \cos^2 x$ Dependent on 1st M	
		i	$\Rightarrow nI_n = (n-1)I_{n-2} \Rightarrow I_n = \frac{n-1}{n} I_{n-2}$	A1	www AG	
Examiner's Comments There were many good complete solutions seen. Most used the expected split of $\cos^{n-1} x \cos x$ as the starting point for their attempt at integration by parts. It was common for sign errors to occur at the first stage. Some tried an approach involving $\cos^{n-2} x \cos^2 x$ but often failed to get very far, though some reached the correct conclusion. Another common error involved starting with a $\cos^n x \times 1$ split; again such attempts quickly came to a halt. The problem with the early sign error meant that a second "error" had to be made to put it right, either inadvertently or deliberately. This of course is of no benefit as the original error will have been penalised. It appeared that many candidates, however, on realising that the result was not going to come out, checked back over their work and found their error.						
		ii	$I_1 = 1$	B1	For I_1 soi	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$I_{11} = \frac{10}{11} I_9 = \dots = \frac{10}{11} \cdot \frac{8}{9} \cdot \frac{6}{7} \cdot \frac{4}{5} \cdot \frac{2}{3} I_1$	M1	Use of (i) to give product of 5 fractions	
		ii	$\Rightarrow I_{11} = \frac{3840}{10395} = \frac{256}{693} \text{ oe}$	A1	Correct fraction	
					Examiner's Comments	
					A significant majority of candidates obtained a correct answer for this part, even though the answer to part (i) was not obtained. Common errors were to give $I_1 = 81$ or to have a careless slip in one of the fractions.	
			Total	8		



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Question			Answer/Indicative content	Marks	Part marks and guidance	
77		i	identity $0 + 0i$	B1	Or '0'	
		i	order 25	B1	<u>Examiner's Comments</u> This question was generally answered well, particularly when it came to stating the identity. A common error with the order of the group was to say that it was 5 rather than 25.	
		ii	$3 + i$	B1	<u>Examiner's Comments</u> A significant number of candidates, erroneously, gave the answer as $-2 - 4i$, failing to realise that this was not an element of the group. A smaller number, for some reason, thought that the inverse had to be the complex conjugate.	
		iii	$5(a + bi) = 5a + 5bi = 0 + 0i$	M1	Shows 5 times any element equals e	
		iii	every non-zero element has order 5 or 25	M1	Attempt to show that order $\neq 2,3,4$	Must consider all(non-zero) elements



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	So order is 5	A1	Argument is convincing, exhaustive and conclusive. <u>Examiner's Comments</u> This part of the question allowed candidates to demonstrate the thoroughness of their mathematical arguments. Some, however tended not to realise that $5(a + bi) = e$, whilst a necessary condition, was not a sufficient one without giving further reasoning. The strongest candidates either used Lagrange's theorem to help them complete the demonstration, or were exhaustive in their exploration of the multiples of all possible $a + bi$, considering $a \neq 0$ or $b \neq 0$ (or both).	
			Total	6		



Question			Answer/Indicative content	Marks	Part marks and guidance	
78		i	$wa^2 = waa = a^3 wa = a^3 a^3 w$	M1	Use $wa = a^3 w$ to simplify	
		i	$= a^4 a^2 w = ea^2 w$	B1	Use $a^4 = e$ (oe) in either proof	
		i	$= a^2 w$	A1	Complete argument AG	
		i	Either result $\Rightarrow wa^3 = a^3 wa^2$	M1		
		i	$= a^3 a^2 w$	M1		
		i	$= eaw = aw$	A1	AG	
					<u>Examiner's Comments</u> The most elegant answers started from the given identity and worked from there to the new identity in clear steps with reference to the use of $a^4 = e$ or $w^2 = e$. Other well-written arguments worked from one side of the new identity to the other. Where candidates started with the identity to be shown and operated on both sides of the identity till both sides were shown to equate, they still gained the marks on this occasion. However, this practice should be discouraged.	
Copyright © 2024 Exam Papers Practice		ii	$(aw)^2 = (aw)(aw)$	M1	for squaring any of elements	
		ii	$= awwa^3 = aea^3 = a^4 = e$ so order 2	M1	for attempt to simplify to e	
		ii	$(a^2 w)(a^2 w) = a^2 wwa^2 = a^2 ea^2 = a^4 = e$ so order 2	A1	for at least two squared elements shown equal to e	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$(a^3 w)(a^3 w) = a^3 wwa = a^3 ea = a^4 = e$ so order 2	A1	for complete argument <u>Examiner's Comments</u> There were a few candidates who both here – and sometimes in part (i) – falsely assumed commutativity. Most others recognised what was required and made a good attempt to simplify the squared terms they produced.	
		iii	$\{e, a^2, w, a^2 w\}$	B1		Condone equivalents
		iii	$\{e, a^2, aw, a^3 w\}$	B1		
		iii	$a^2, w, aw, a^2 w, a^3 w$ all of order 2	M1	Consider orders Or considers form $\{e, x, y, xy\}$ where x, y order 2	
		iii	so not cyclic as no element of order 4 in either	A1	Dep on both groups correct <u>Examiner's Comments</u> While many candidates recognised the need to use only elements of order 2 (plus e) in their subgroups, some of these had clearly not considered ensuring that their subset was closed. So it was common to see sets such as $\{e, aw, aw^2, aw^3\}$ alongside one correct cyclic subgroup. The strongest candidates often demonstrated a familiarity not only with general group theory, but also with the structure of both groups of order 4.	Condone 'no generator' or 'Klein (V) group' in place of 'no element of order 4'
			Total	14		



Question			Answer/Indicative content	Marks	Part marks and guidance	
79			$3(2 \times 3^n - 1) + 2$	B1 M1 A1 B1	Show clearly result true for $n = 1$, accept $(u_1) = 2 \times 3 - 1 = 5$ Substitute for u_n in recurrence relation Establish correct result for u_{n+1} convincingly Clear statement of induction conclusion, provided 1st 3 marks earned Examiner's Comments Some candidates failed to show clearly that the result is true for $n = 1$, while others started the process at $n = 2$. Most were able to show that using the given form in the recurrence relation set up the induction process correctly. A significant number of candidates did not give a clear explanation of the induction process, so the last mark was often lost.	
			Total	4		

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